

Natural Language Processing

06 – Multimodal Models

Presentation Agenda

- Introduction
 - General concepts;
 - Challenges;
- Vision Transformers
 - Overview;
- CLIP
- Flamingo
- LLaVA
- Native Multimodal Models

General Concepts and Taxonomy

Multimodal Machine Learning: A Survey and Taxonomy

Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency

Abstract—Our experience of the world is multimodal - we see objects, hear sounds, feel texture, smell odors, and taste flavors. Modality refers to the way in which something happens or is experienced and a research problem is characterized as *multimodal* when it includes multiple such modalities. In order for Artificial Intelligence to make progress in understanding the world around us, it needs to be able to interpret such multimodal signals together. Multimodal machine learning aims to build models that can process and relate information from multiple modalities. It is a vibrant multi-disciplinary field of increasing importance and with extraordinary potential. Instead of focusing on specific multimodal applications, this paper surveys the recent advances in multimodal machine learning itself and presents them in a common taxonomy. We go beyond the typical early and late fusion categorization and identify broader challenges that are faced by multimodal machine learning, namely: representation, translation, alignment, fusion, and co-learning. This new taxonomy will enable researchers to better understand the state of the field and identify directions for future research.

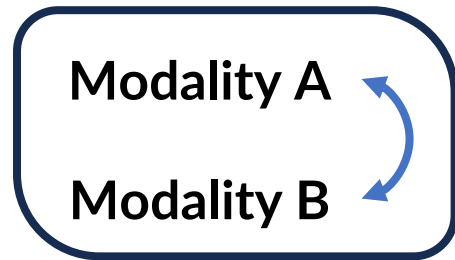
Index Terms—Multimodal, machine learning, introductory, survey.



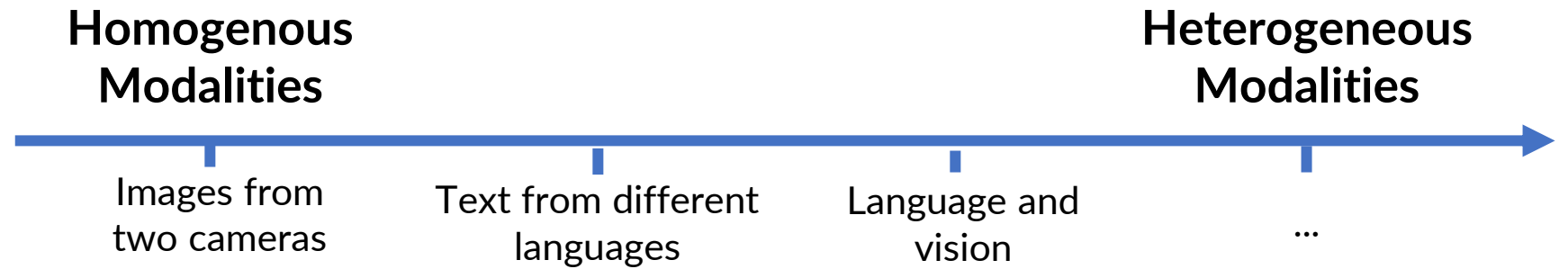
Baltrušaitis, T., Ahuja, C. and Morency, L.P., 2018. Multimodal machine learning: A survey and taxonomy. *IEEE transactions on pattern analysis and machine intelligence*, 41(2), pp.423-443.

General Concepts and Taxonomy

- Multimodal is the science of **heterogeneous** and **interconnected** data



Examples:



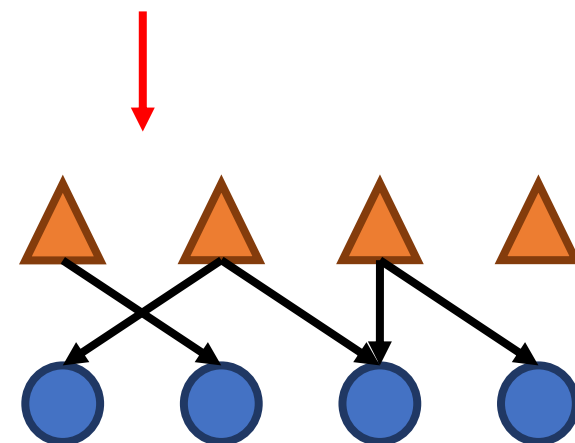
General Concepts and Taxonomy

- Multimodal is the science of **heterogeneous** and **interconnected** data

- Connections
- Cross-modal Interactions

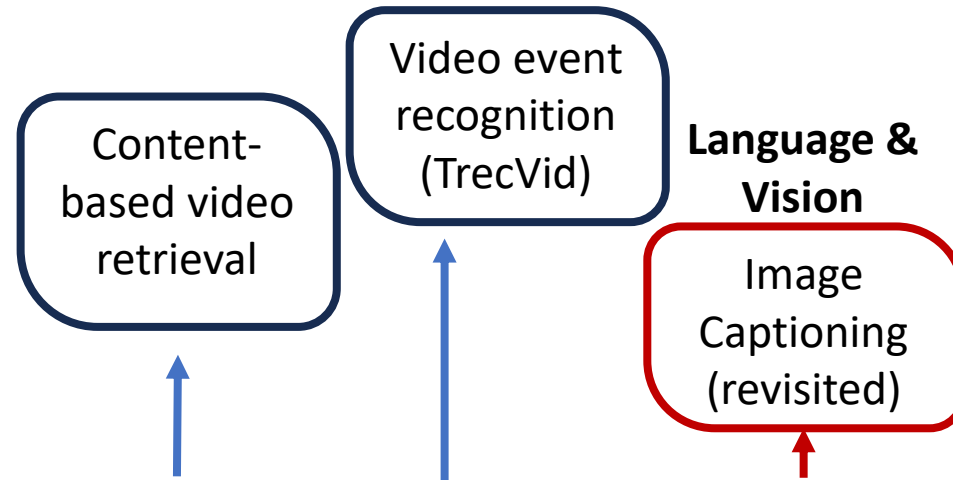
Modality A

Modality B

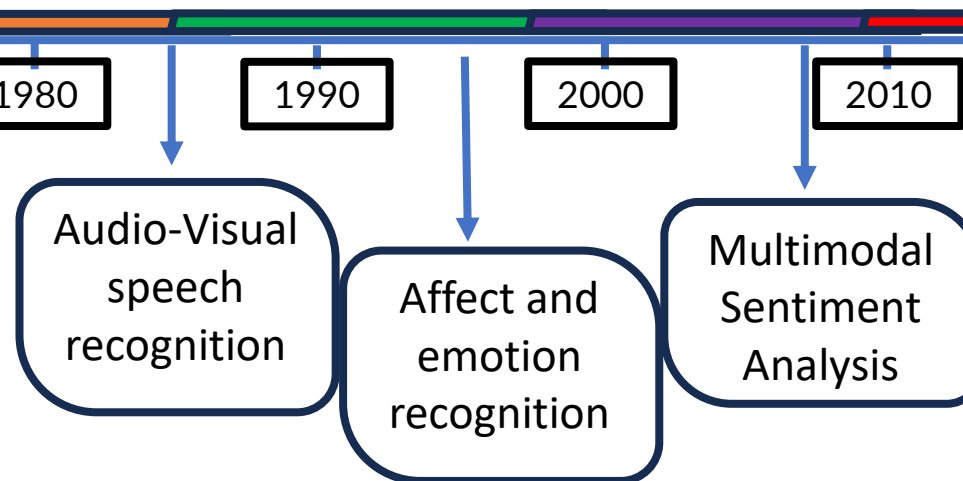


Research Tasks History

- The “behavioral” era;
- The “computational” era;
- The “interaction” era;
- The “deep learning” era;



McGurk Effect



Multimodal Machine Learning

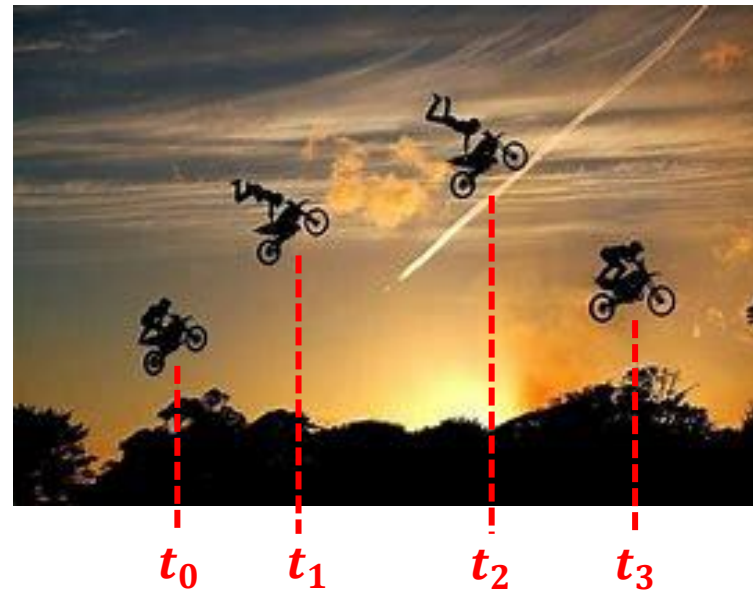


Modality A

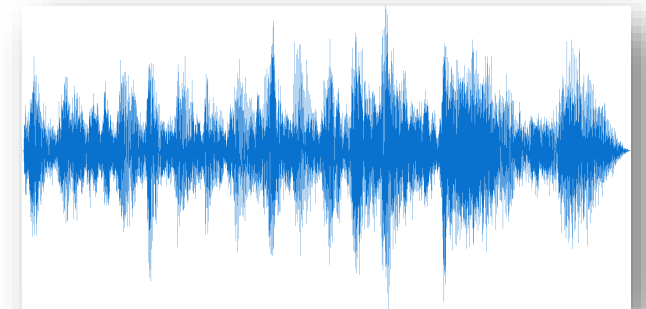
Modality C

Multimodal machine learning has several challenges...

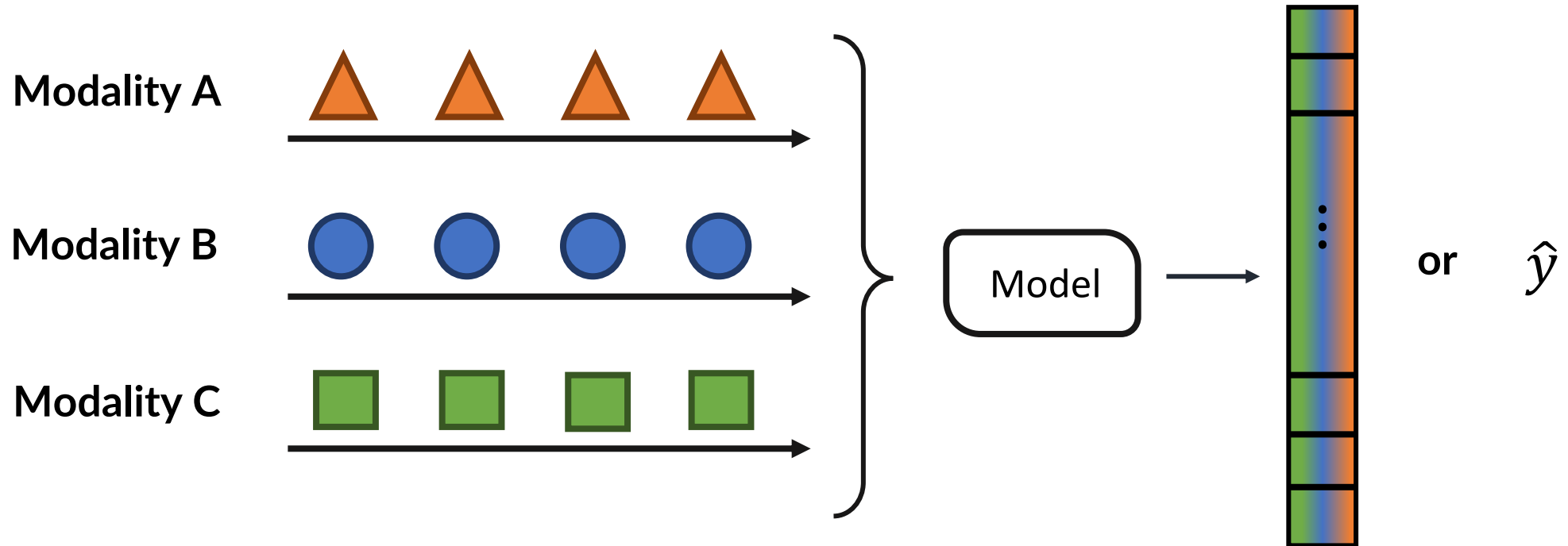
Modality B



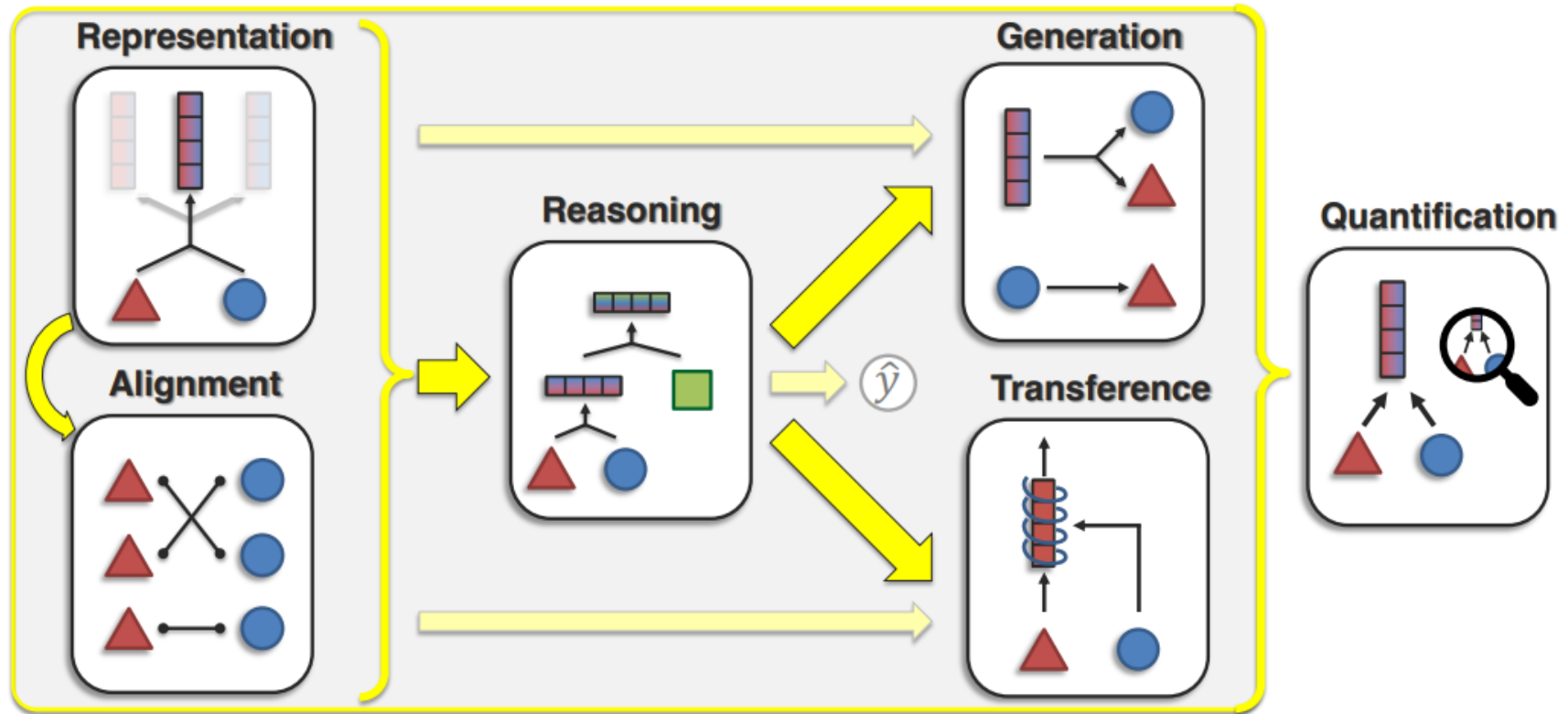
Modality D



Multimodal Machine Learning



Core Technical Challenges



Core Technical Challenges

Challenge	Subchallenge	Approaches & key examples
Representation (Section 3)	Fusion (Section 3.1)	Abstract [148, 396] & raw [30, 272] fusion
	Coordination (Section 3.2)	Strong [95, 268] & partial [352, 407] coordination
	Fission (Section 3.3)	Modality-level [121, 337] & fine-grained [1, 58] fission
Alignment (Section 4)	Discrete connections (Section 4.1)	Local [71, 129] & global [182] alignment
	Continuous alignment (Section 4.2)	Warping [115, 132] & segmentation [314]
	Contextualization (Section 4.3)	Joint [180], cross-modal [120, 207] & graphical [385]
Reasoning (Section 5)	Structure modeling (Section 5.1)	Hierarchical [20], temporal [376], interactive [211] & discovery [260]
	Intermediate concepts (Section 5.2)	Attention [379], discrete symbols [18, 350] & language [140, 404]
	Inference paradigm (Section 5.3)	Logical [107, 318] & causal [6, 243, 390]
	External knowledge (Section 5.4)	Knowledge graphs [111, 417] & commonsense [253, 402]
Generation (Section 6)	Summarization (Section 6.1)	Extractive [62, 346] & abstractive [177, 250]
	Translation (Section 6.2)	Exemplar-based [153, 170] & generative [10, 146, 273]
	Creation (Section 6.3)	Conditional decoding [79, 245, 414]
Transference (Section 7)	Cross-modal transfer (Section 7.1)	Tuning [271, 341], multitask [194, 303] & transfer [208]
	Co-learning (Section 7.2)	Representation [149, 398] & generation [262, 321]
	Model Induction (Section 7.3)	Co-training [40, 87] & co-regularization [308, 387]
Quantification (Section 8)	Heterogeneity (Section 8.1)	Importance [103, 252], bias [119, 258] & noise [214]
	Interconnections (Section 8.2)	Connections [5, 49, 327] & interactions [121, 193, 363]
	Learning (Section 8.3)	Generalization [194, 275], optimization [362, 371] & tradeoffs [195]

Liang, Paul Pu, Amir Zadeh, and Louis-Philippe Morency. "Foundations & trends in multimodal machine learning: Principles, challenges, and open questions." *ACM computing surveys* 56.10 (2024): 1-42.

Core Technical Challenges

	CHALLENGES				
APPLICATIONS	REPRESENTATION	TRANSLATION	ALIGNMENT	FUSION	CO-LEARNING
Speech recognition and synthesis					
Audio-visual speech recognition	✓		✓	✓	✓
(Visual) speech synthesis	✓	✓			
Event detection					
Action classification	✓			✓	✓
Multimedia event detection	✓			✓	✓
Emotion and affect					
Recognition	✓		✓	✓	✓
Synthesis	✓	✓			
Media description					
Image description	✓	✓	✓		✓
Video description	✓	✓	✓	✓	✓
Visual question-answering	✓		✓	✓	✓
Media summarization	✓	✓		✓	
Multimedia retrieval					
Cross modal retrieval	✓	✓	✓		✓
Cross modal hashing	✓				✓

An Image is Worth 16x16 Words: Transformers For Image Recognition at Scale

**AN IMAGE IS WORTH 16X16 WORDS:
TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE**

**Alexey Dosovitskiy^{*,†}, Lucas Beyer^{*}, Alexander Kolesnikov^{*}, Dirk Weissenborn^{*},
Xiaohua Zhai^{*}, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer,
Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*,†}**

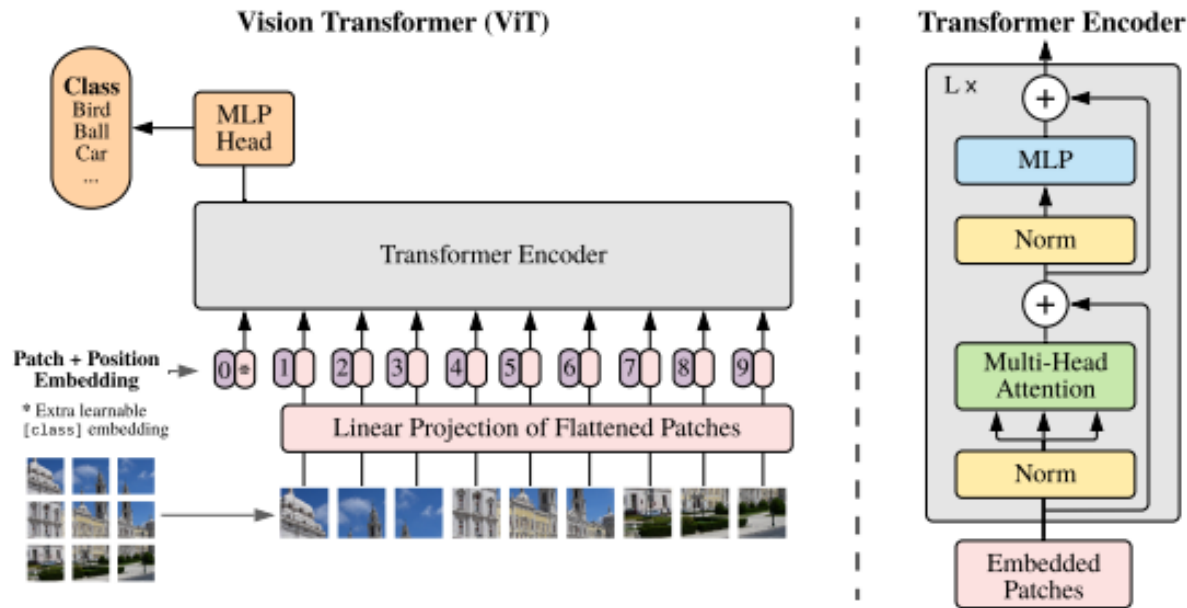
^{*}equal technical contribution, [†]equal advising

Google Research, Brain Team

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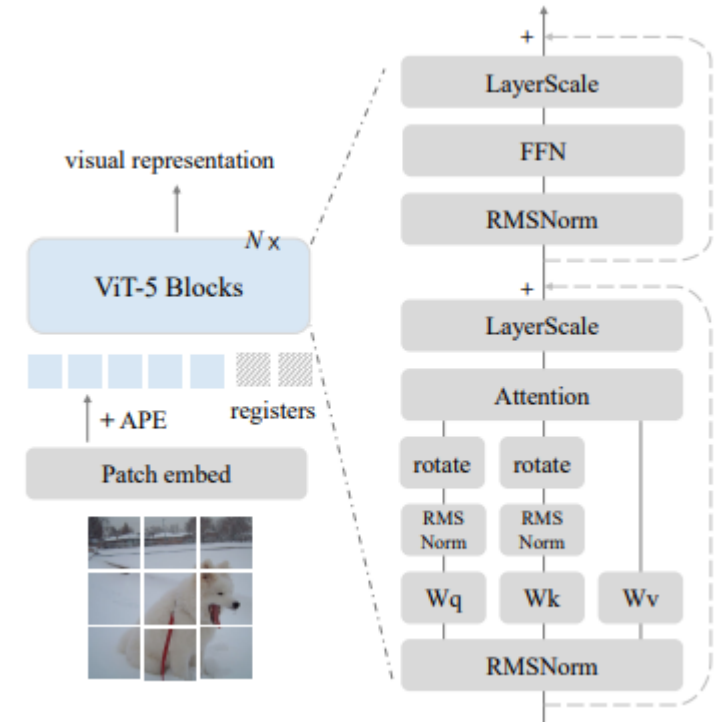
Proposed Approach

a)



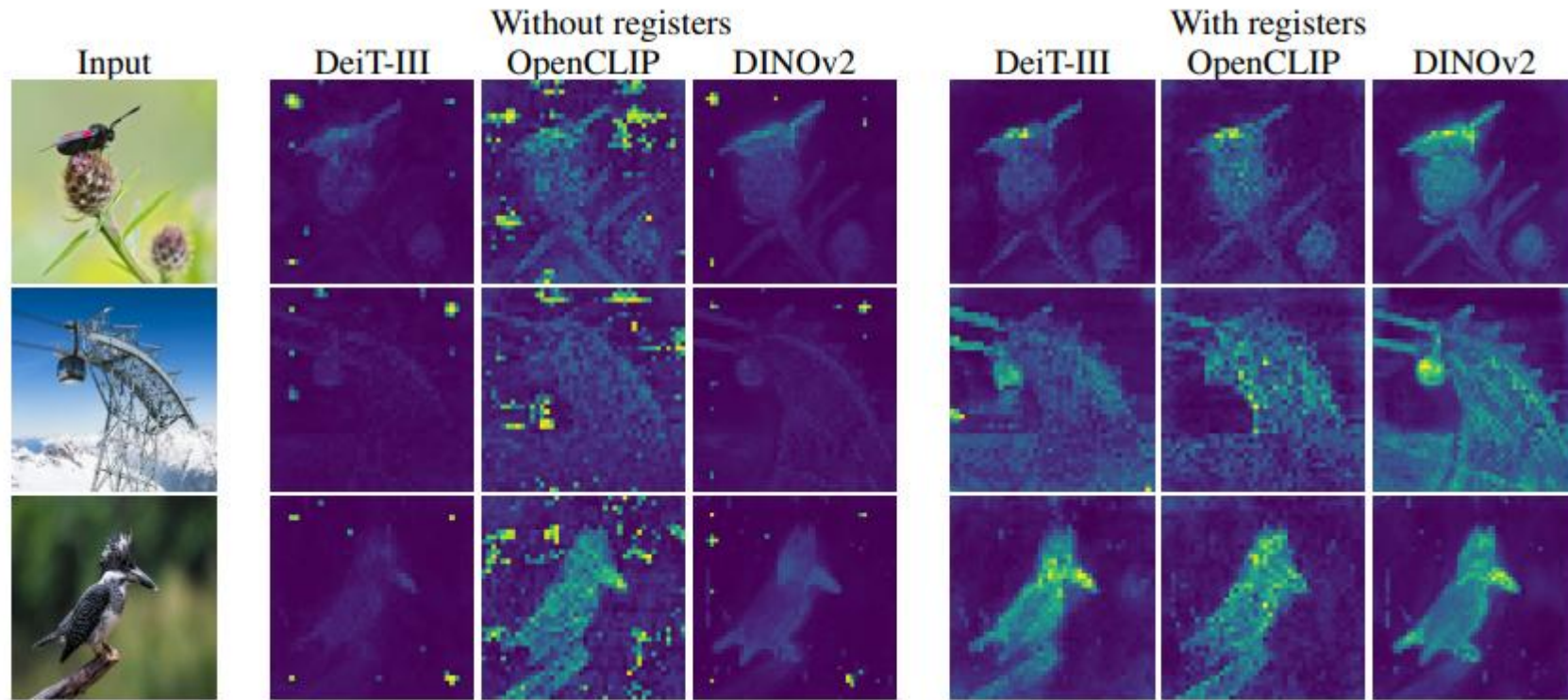
[1] - Dosovitskiy, Alexey. "An image is worth 16x16 words: Transformers for image recognition at scale." *arXiv preprint arXiv:2010.11929* (2020).

b)



[2] - Wang, Feng, et al. "Vit-5: Vision transformers for the mid-2020s." *arXiv preprint arXiv:2602.08071* (2026).

Proposed Approach



Darcet, Timothée, et al. "Vision transformers need registers." *International conference on learning representations*. Vol. 2024. 2024.

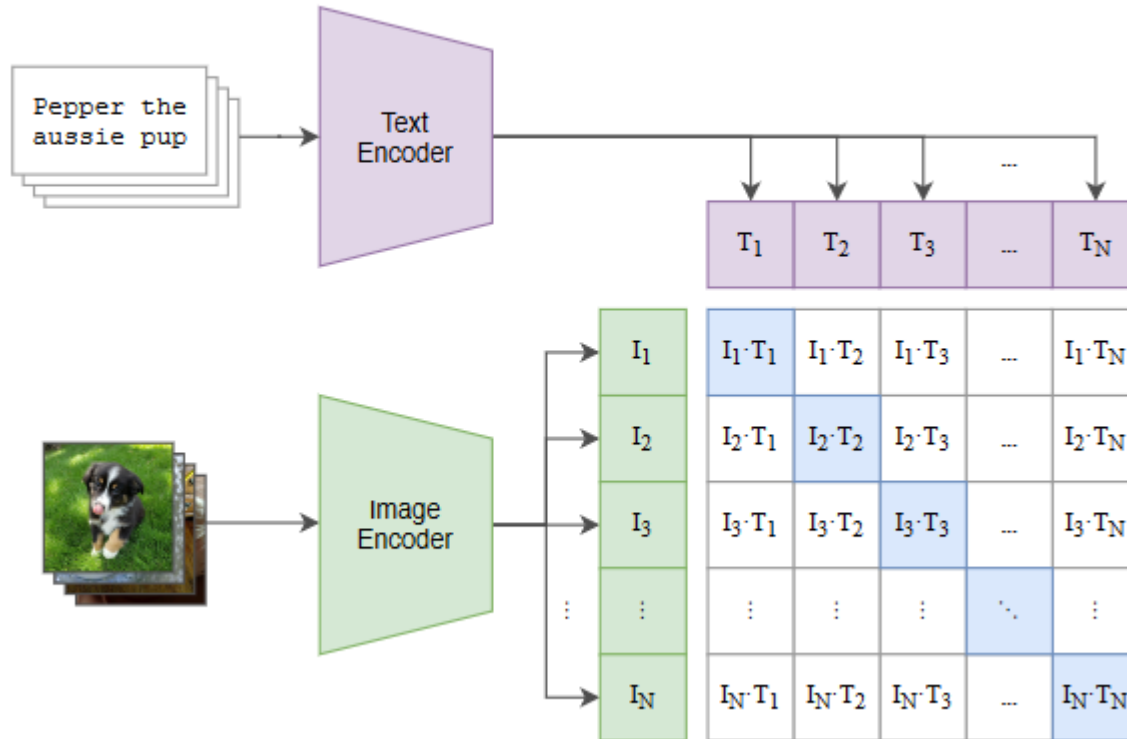
Learning Transferable Visual Models From Natural Language Supervision

Learning Transferable Visual Models From Natural Language Supervision

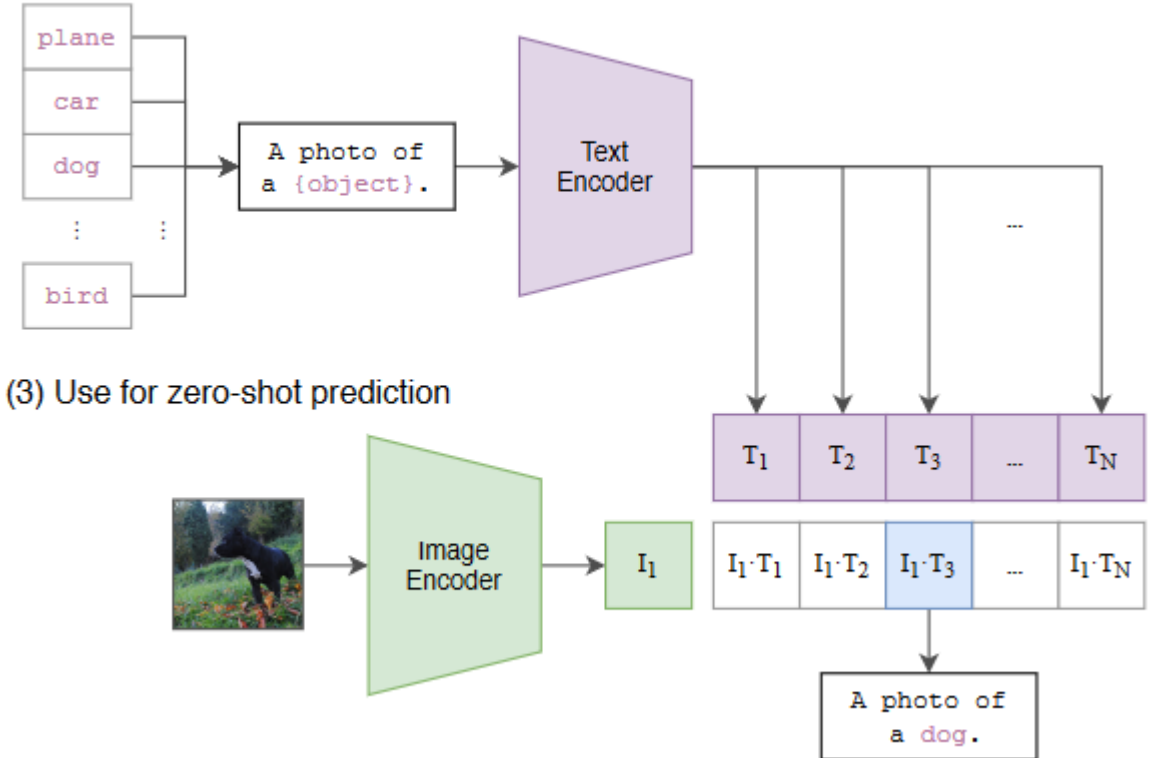
**Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Aspell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹**

CLIP - Proposed Approach

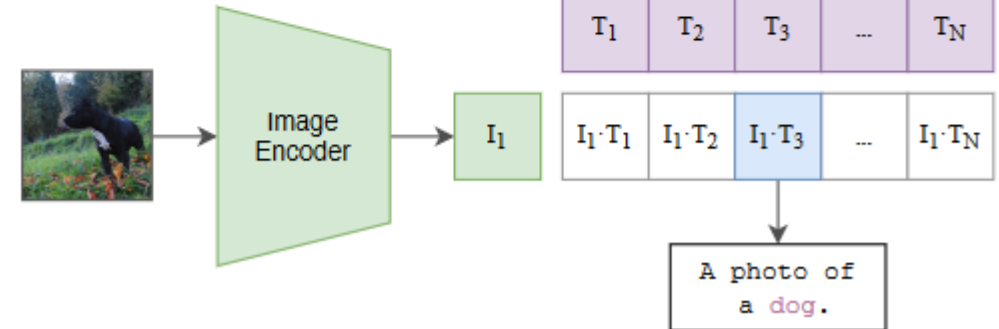
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



CLIP - Proposed Approach

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer (GPT-2 based)
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t            - learned temperature parameter

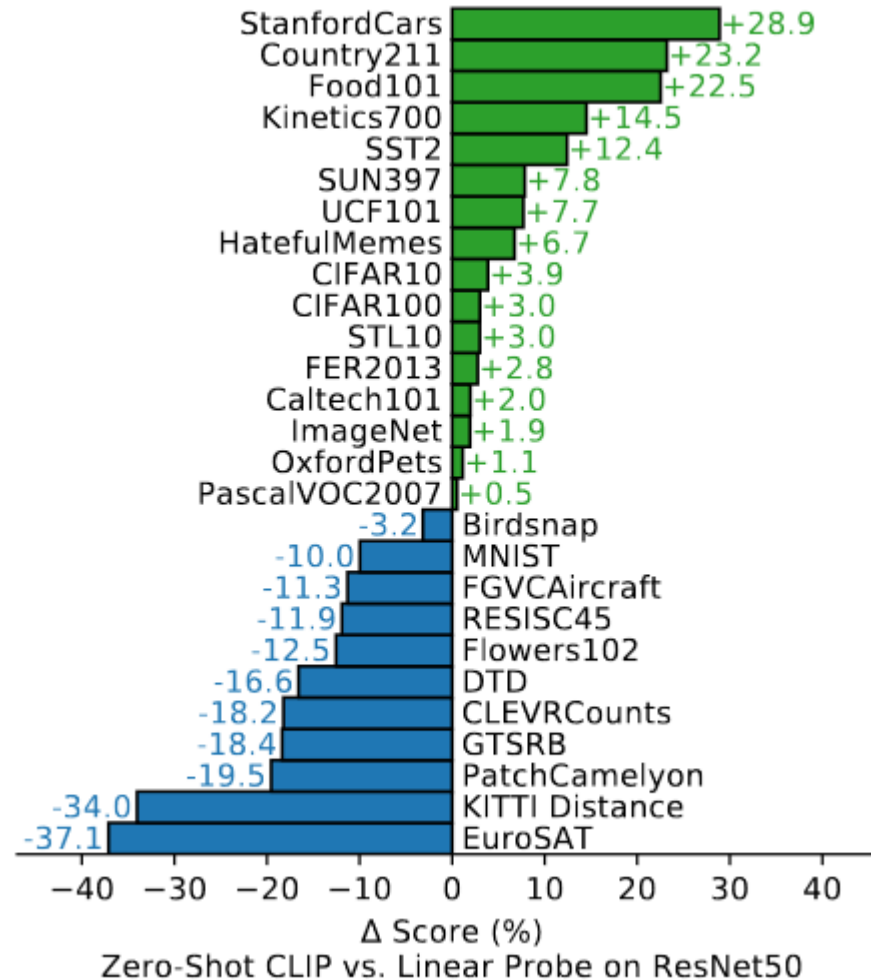
# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

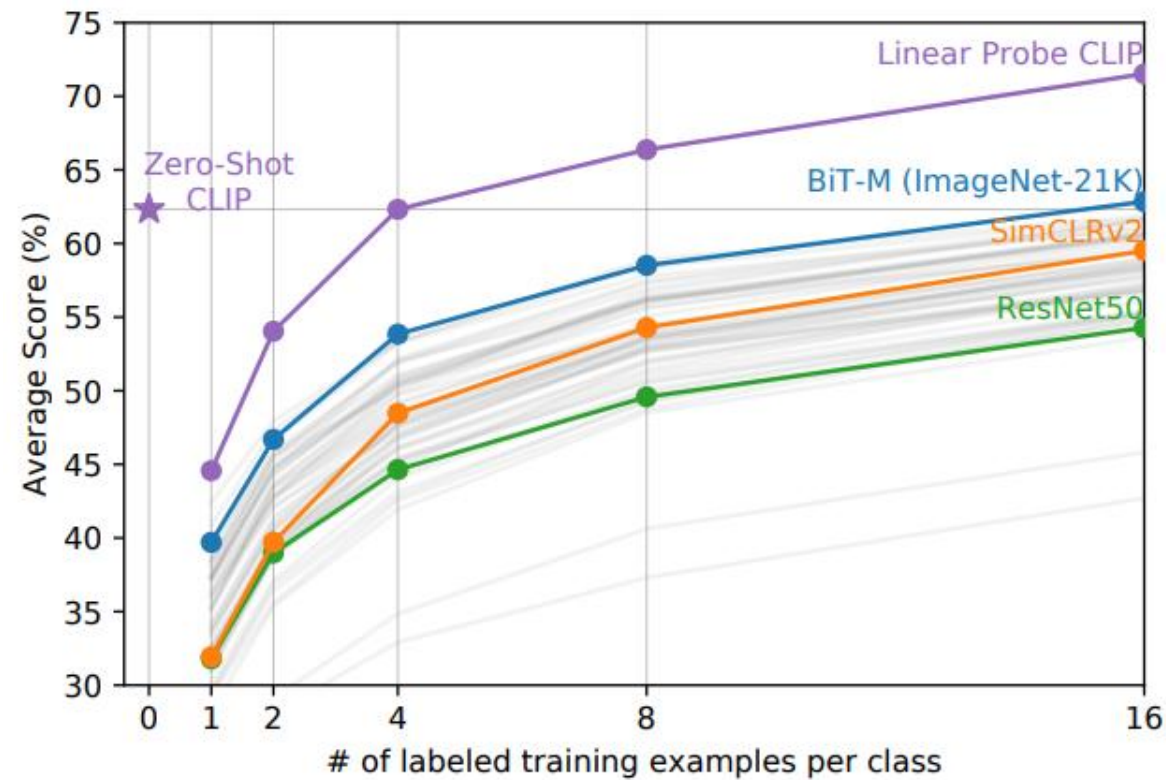
# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

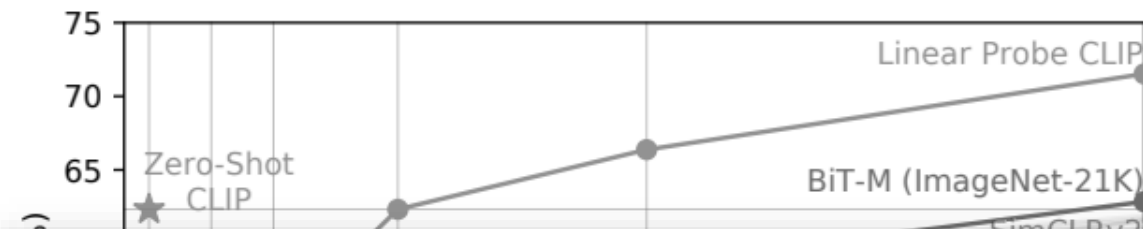
Zero-Shot CLIP vs. Linear Probe on ResNet50



Zero-Shot CLIP



Zero-Shot CLIP

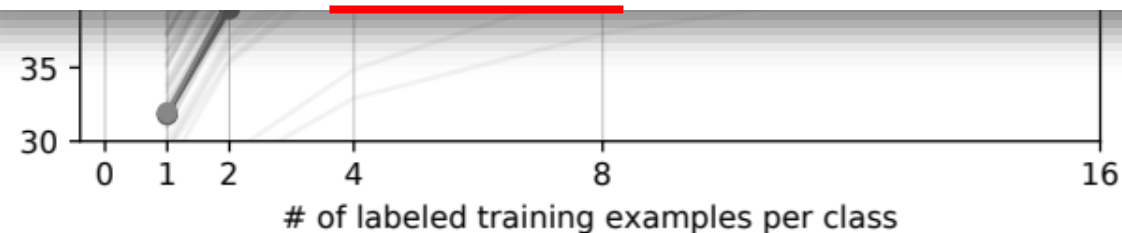


Learning transferable visual models from natural language supervision

A Radford, JW Kim, C Hallacy... - International ..., 2021 - proceedings.mlr.press

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since ...

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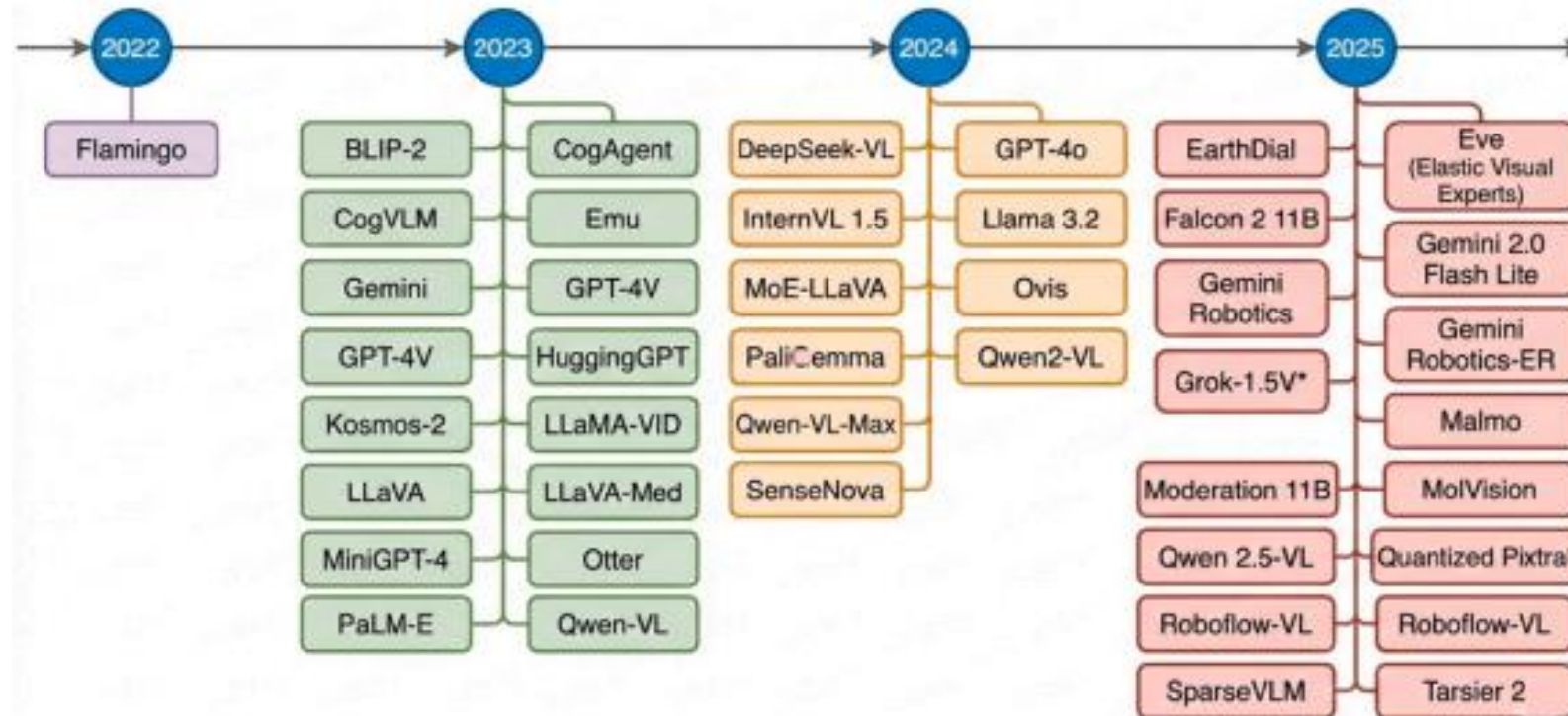
Flamingo: a Visual Language Model for Few-Shot Learning

Alayrac, Jean-Baptiste, et al. "Flamingo: a visual language model for few-shot learning." *Advances in neural information processing systems* 35 (2022): 23716-23736.

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Why Flamingo?

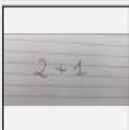
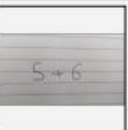
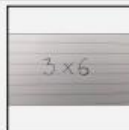
Progression of Vision-Language Models (2022–2025)



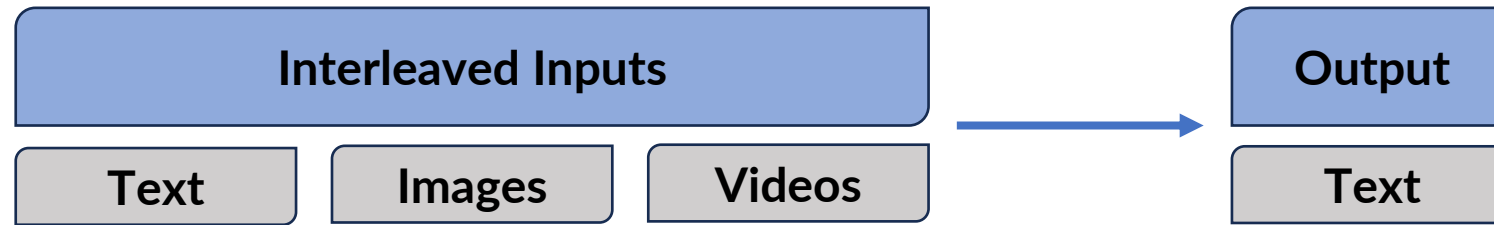
Rahman, Arifur. "A systematic review of vision language models: Comprehensive analysis of architectures, applications, datasets and challenges towards robust multimodal intelligence." *Array* (2026): 100739.

Motivation

- The few-shot dream:
 - Short instruction;
- Multimodal models like CLIP have shown promising zero-shot performance, but they lack the ability to **generate language**;

Input Prompt						Completion
	This is a chinchilla. They are mainly found in Chile.		This is a shiba. They are very popular in Japan.		This is	a flamingo. They are found in the Caribbean and South America.
	What is the title of this painting? Answer: The Hallucinogenic Toreador.		Where is this painting displayed? Answer: Louvres Museum, Paris.		What is the name of the city where this was painted? Answer:	Arles.
	Output: "Underground"		Output: "Congress"		Output:	"Soulomes"
	2+1=3		5+6=11			3x6=18
	Output: A propaganda poster depicting a cat dressed as French emperor Napoleon holding a piece of cheese.		Output: A pink room with a flamingo pool float.		Output:	A portrait of Salvador Dali with a robot head.

Proposed approach

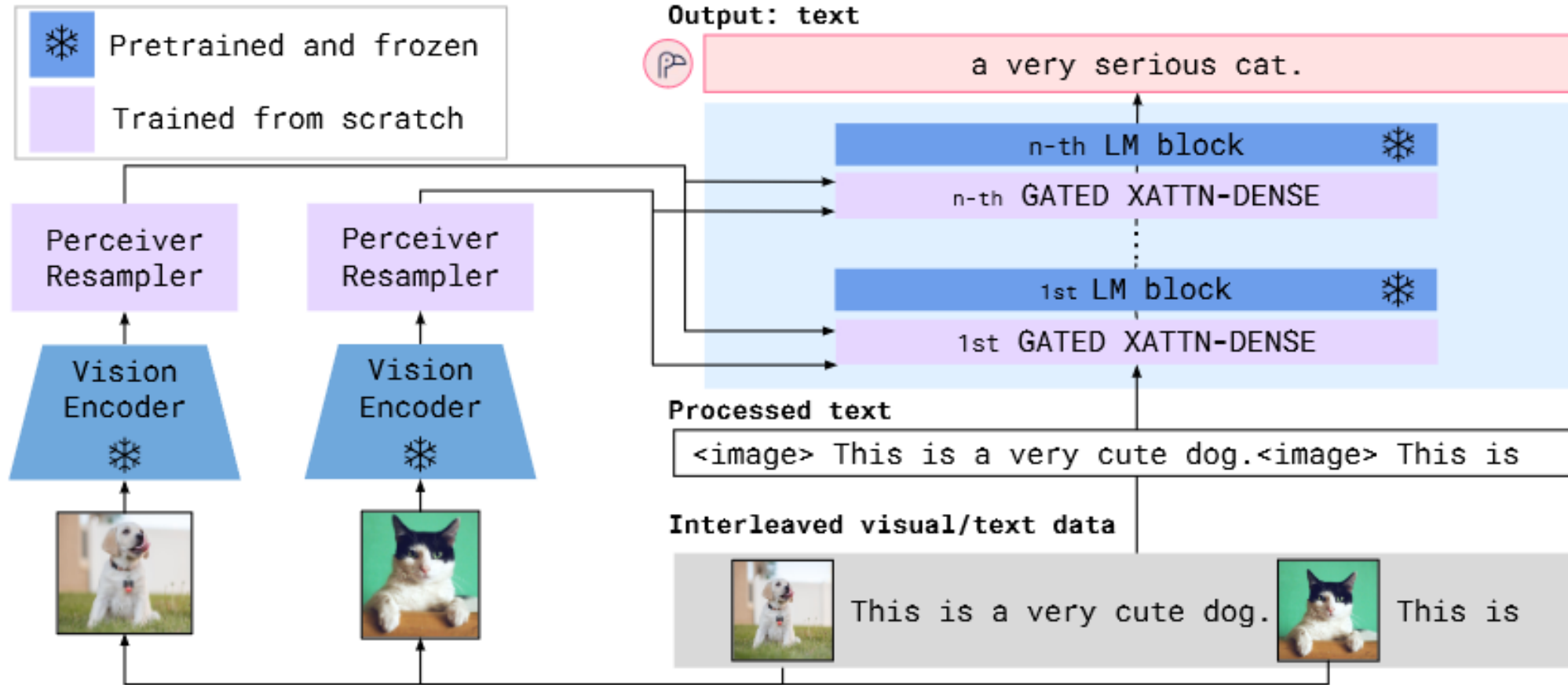


- **Open-ended tasks:**
 - Visual question answering;
 - Cross-modal Interactions;
- **Close-ended tasks:**
 - Classification;

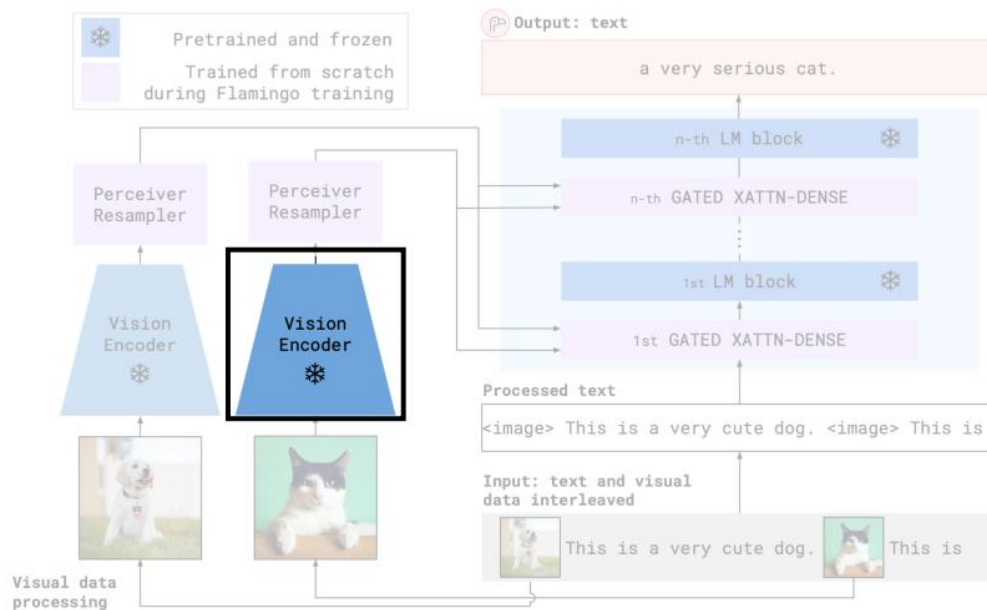
Goal 1: Leverage **pretrained models** to save compute

Goal 2: **Bridge** pretrained models

Architecture



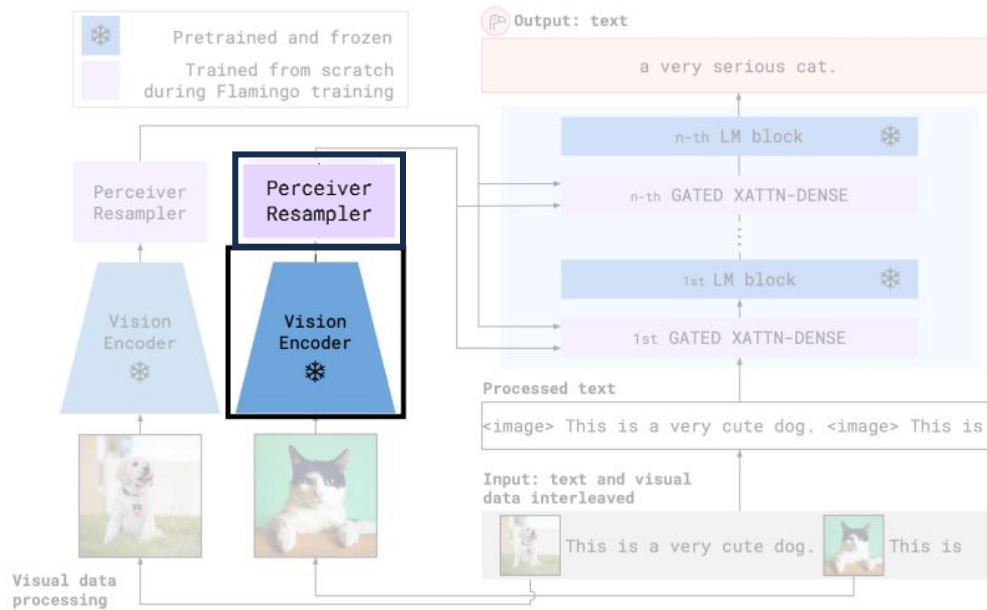
Architecture



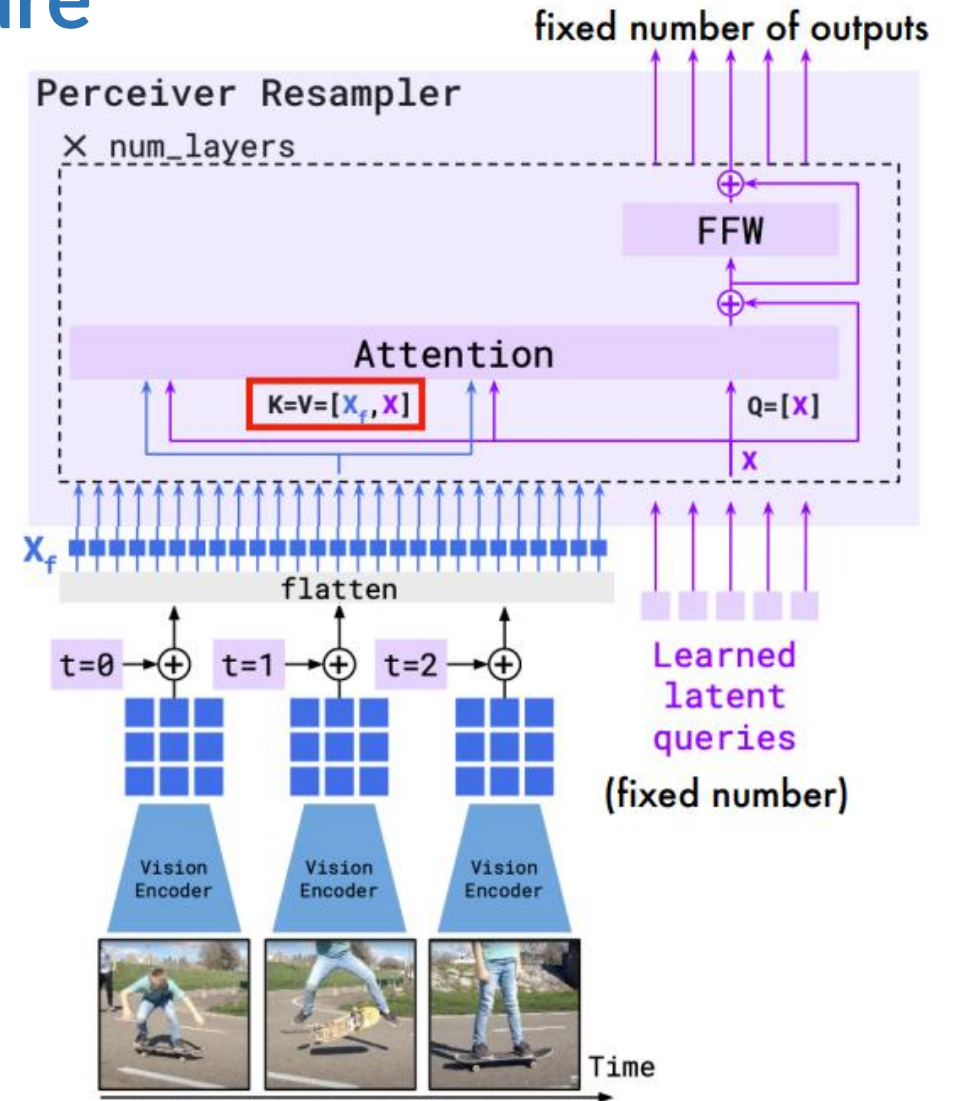
- **Normalizer-Free ResNet (NFNet):**
 - Pretrained as dual encoder using contrastive loss;
 - BERT is used for the text encoder;
 - Input resolution of 288x288 pixels;
 - Embedding size of 1376;
- **Trained on two (image, text) datasets, combined by accumulation:**
 - ALIGN (1.8 billion) – noisy
 - LTIP (312 million) – cleaner, longer descriptions

Vision encoder is frozen after pretraining

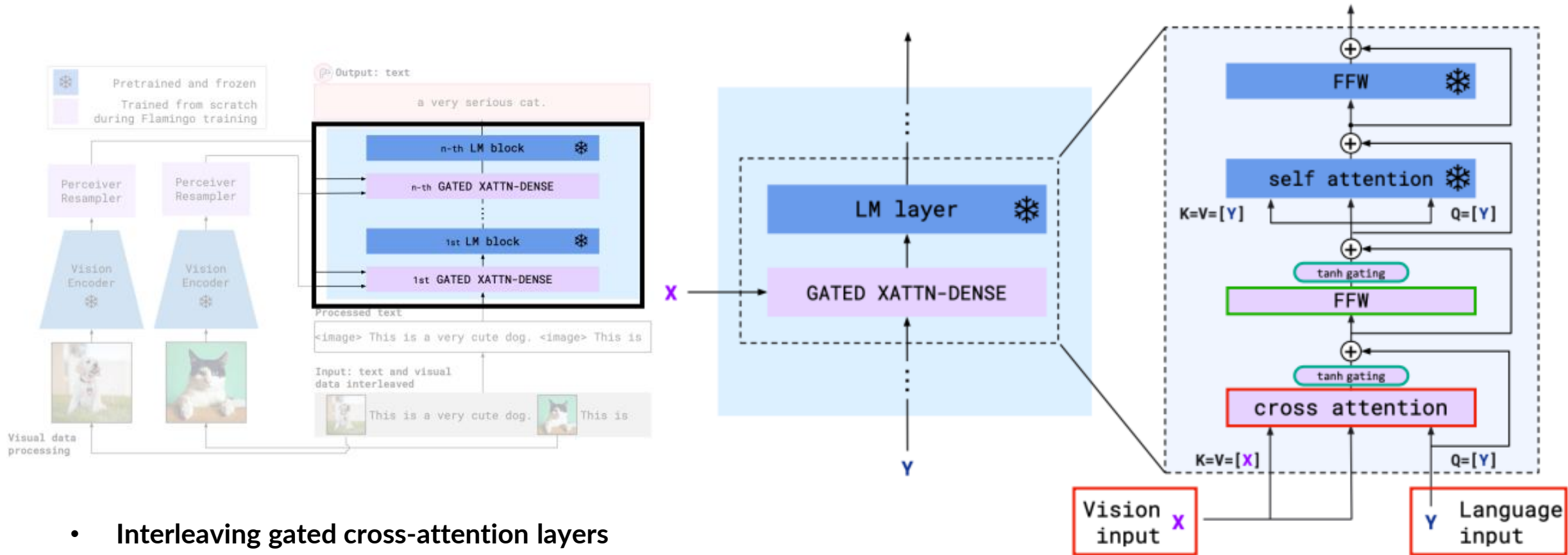
Architecture



- **Goal:** outputs a fixed number of visual tokens to limit complexity

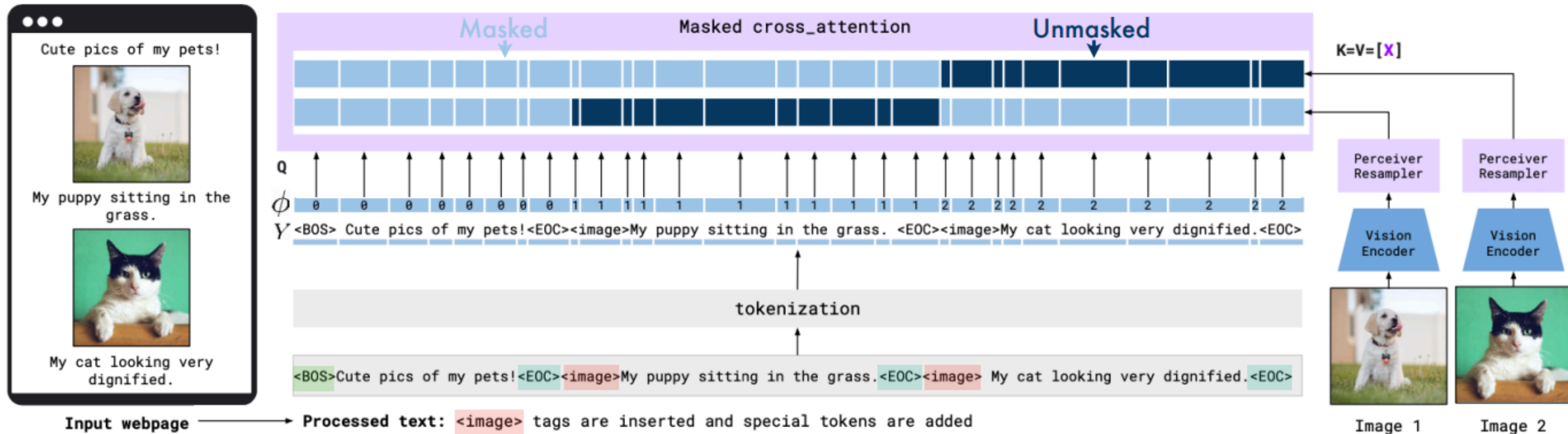


Architecture



- **Interleaving gated cross-attention layers**
 - Frozen chinchila layers;
 - Gates to preserve original language model behaviour at initialization

Per-image/video attention masking



Flamingo - Training Data

Data-sources

- Training data:
 - **Image-text Pairs;**
 - **Video-text Pairs;**
 - **Webpage;**



Image-Text Pairs dataset
[N=1, T=1, H, W, C]



Video-Text Pairs dataset
[N=1, T>1, H, W, C]



Multi-Modal Massive Web (M3W) dataset
[N>1, T=1, H, W, C]



- **Image-Text Pairs data:**
 - **ALIGN** (1.8 billion) – noisy;
 - 12.4 mean text tokens per image
 - **LTIP** (312 million) – cleaner, longer descriptions;
 - 20.5 text tokens per image on average
- **320x320 Resolution**
- **$L = 32/64$;**

Flamingo - Training Data

Data-sources

- Training data:
 - **Image-text** Pairs;
 - **Video-text** Pairs;
 - **Webpage**;



Image-Text Pairs dataset
[N=1, T=1, H, W, C]



Video-Text Pairs dataset
[N=1, T>1, H, W, C]



Multi-Modal Massive Web (M3W) dataset
[N>1, T=1, H, W, C]



- **Video-Text Pairs** data:
 - **VTP** dataset (27 million) – short videos;
 - 22 seconds on average;
 - **320x320 resolution**;
 - $L = 32$;
 - $T=8$;

Flamingo - Training Data

Data-sources

- Training data:
 - **Image-text** Pairs;
 - **Video-text** Pairs;
 - **Webpage**;



Image-Text Pairs dataset
[N=1, T=1, H, W, C]



Video-Text Pairs dataset
[N=1, T>1, H, W, C]



Multi-Modal Massive Web (M3W) dataset
[N>1, T=1, H, W, C]



- **Webpage data:**
 - **MultiModal Massive Web (M3W);**
 - 43 million webpages;
 - **185 million** images;
 - **182 GB** of text;
 - **320x320 resolution;**
 - **$L = 256$;**

Flamingo - Training Objective

$$\sum_{m=1}^M \lambda_m \cdot \mathbb{E}_{(x,y) \sim \mathcal{D}_m} \left[- \sum_{l=1}^L \log p(y_l | y_{<l}, x_{\leq l}) \right]^*$$

$\mathcal{D}_m$ ***m*-th dataset**

λ_m positive scalar weight for the ***m*-th dataset**

- The same **accumulation strategy** is used
 - gradient batches for each dataset then weight it by λ

Task adaptation with few-shot in-context learning

- Evaluate ability to **rapidly adapt** to new tasks

Examples of (Image, text) or (vídeo, text)

+

Visual query

Vision to Text tasks (input=vision, output=text)

Support examples



A cat wearing sunglasses.



Elephants walking in the savanna.



Query

Output:

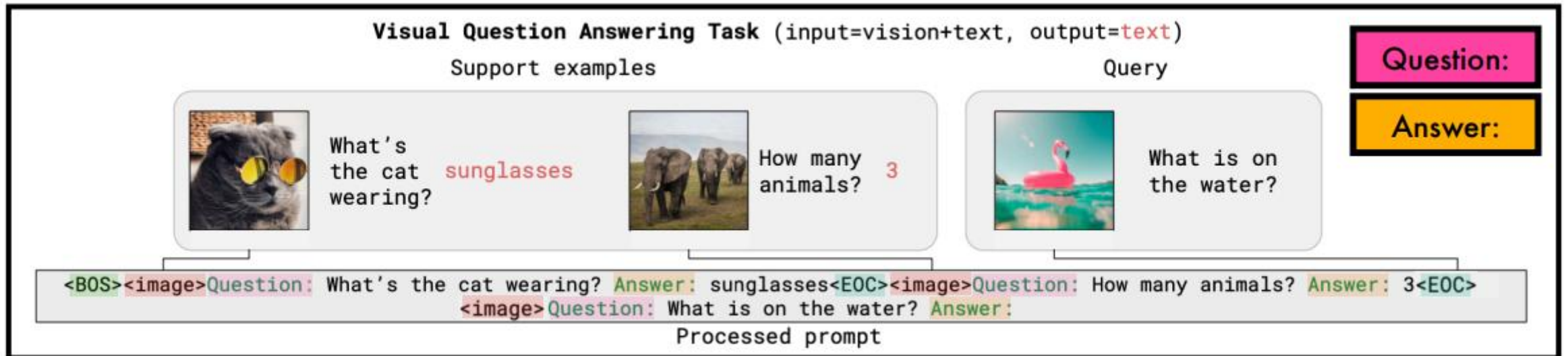
<BOS><image>Output: A cat wearing sunglasses.<EOC><image>Output: Elephants walking in the savanna.<EOC><image>Output:

Processed prompt

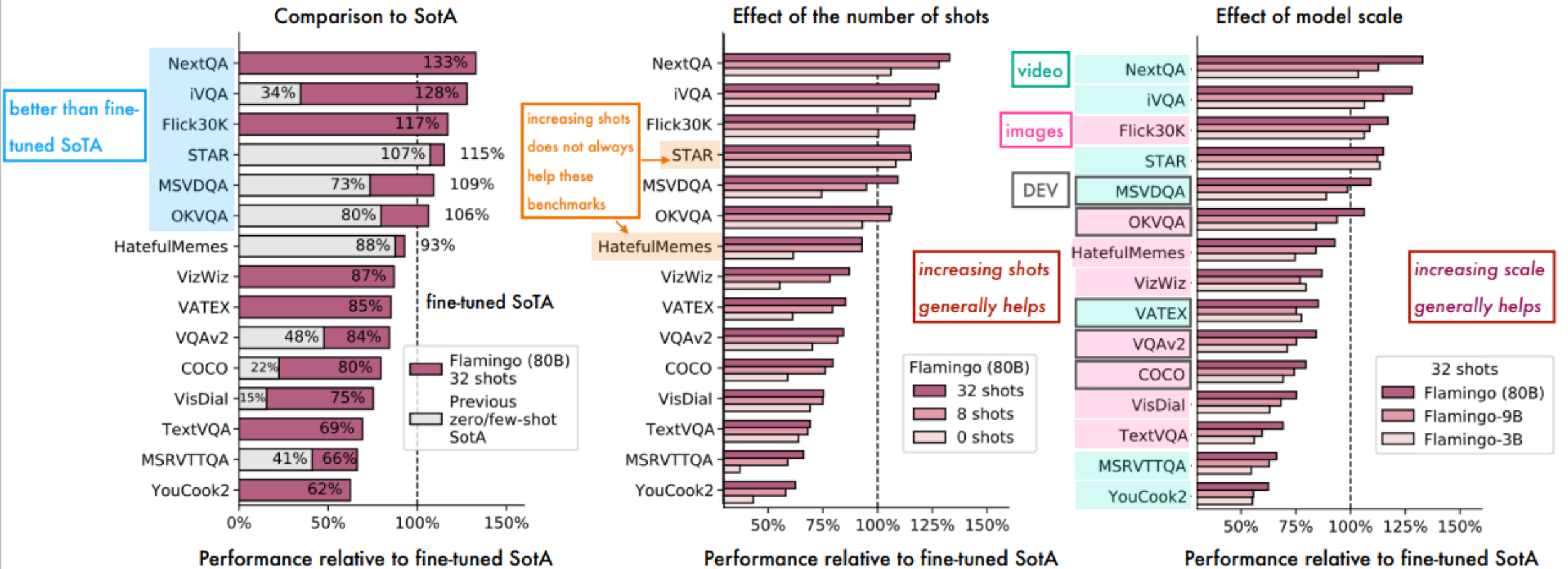
Task adaptation with few-shot in-context learning

- Evaluate ability to **rapidly adapt** to new tasks

Examples of (Image, text) or (vídeo, text) + Visual query



Few-Shot comparison



Qualitative Results

Input Prompt	 Question: What do you think the capacities of these are? Answer:	 Question: What is odd about this image? Explain why it is unusual. Answer:	 Question: What country is this? Why do you think so? Answer:
Completion	The floppy disk is 1.44MB and the CD is 700MB.	The image is odd because the elephant is in the back of the truck. It is unusual because elephants are not usually transported in the back of a truck.	It is Canada. I think so because the flag is the Canadian flag.

Qualitative Results

Input Prompt	 Question: What nutrients is the fruit in the image rich in? Answer:	 Question: What is the purpose of this? Answer:	 Question: What season is this? Explain why. Answer:	 Question: What is the title of the book? Answer:	 The ingredients of this dish are	 The soundtrack includes
Completion	Vitamin C, potassium, and fiber.	To keep the mosquitoes out of the bed.	This is winter season. The Christmas tree and the presents are the main reasons why it is winter.	The House Book.	: beef, potatoes, carrots, broccoli, and lemon.	the songs "Let It Go" and "For the First Time in Forever" by Tony Award® winner Idina Menzel, who plays Elsa [...]

Flamingo limitations

- Builds on pretrained language models, **inheriting their weaknesses**;
- Difícult to few-shot when **text output is long**;
- **Hallucinations** and **ungrounded guesses**;

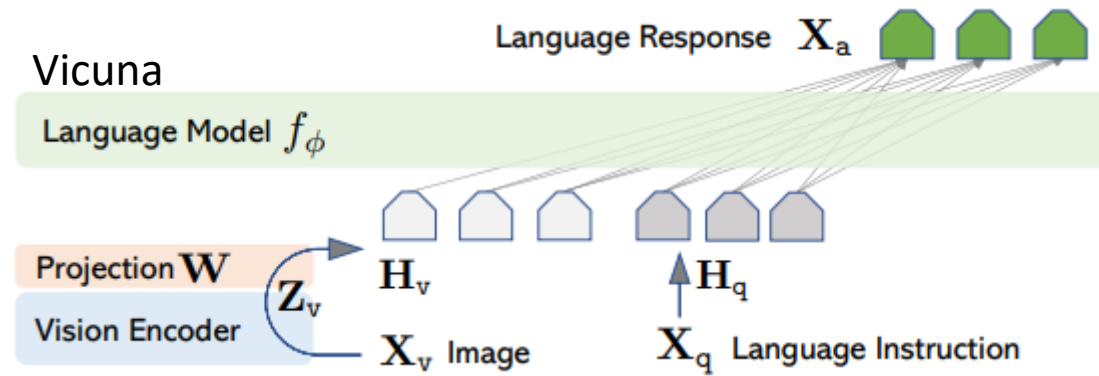
Input Prompt			
	Question: What is on the phone screen? Answer:	Question: What can you see out the window? Answer:	Question: Whom is the person texting? Answer:
Output	A text message from a friend.	A parking lot.	The driver.

Visual Instruction Tuning

Liu, Haotian, et al. "Visual instruction tuning." *Advances in neural information processing systems* 36 (2023): 34892-34916.

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LlaVa - Proposed Approach



$$H_v = W \cdot Z_v, \text{ with } Z_v = g(X_v)$$

LlaVa - Proposed Approach

- ScienceQA contains 21k multimodal multiple choices questions with rich domain diversity across 3 subjects, 26 topics, 127 categories, and 379 skills

Method	Subject			Context Modality			Grade		Average
	NAT	SOC	LAN	TXT	IMG	NO	G1-6	G7-12	
<i>Representative & SoTA methods with numbers reported in the literature</i>									
Human [33]	90.23	84.97	87.48	89.60	87.50	88.10	91.59	82.42	88.40
GPT-3.5 [33]	74.64	69.74	76.00	74.44	67.28	77.42	76.80	68.89	73.97
GPT-3.5 w/ CoT [33]	75.44	70.87	78.09	74.68	67.43	79.93	78.23	69.68	75.17
LLaMA-Adapter [58]	84.37	88.30	84.36	83.72	80.32	86.90	85.83	84.05	85.19
MM-CoT _{Base} [60]	87.52	77.17	85.82	87.88	82.90	86.83	84.65	85.37	84.91
MM-CoT _{Large} [60]	95.91	82.00	90.82	95.26	88.80	92.89	92.44	90.31	91.68
<i>Results with our own experiment runs</i>									
GPT-4 [†]	84.06	73.45	87.36	81.87	70.75	90.73	84.69	79.10	82.69
LLaVA	90.36	95.95	88.00	89.49	88.00	90.66	90.93	90.90	90.92
LLaVA+GPT-4 [†] (complement)	90.36	95.50	88.55	89.05	87.80	91.08	92.22	88.73	90.97
LLaVA+GPT-4 [†] (judge)	91.56	96.74	91.09	90.62	88.99	93.52	92.73	92.16	92.53

Table 7: Accuracy (%) on Science QA dataset. Question categories: NAT = natural science, SOC = social science, LAN = language science, TXT = text context, IMG = image context, NO = no context, G1-6 = grades 1-6, G7-12 = grades 7-12. [†]Text-only GPT-4, our eval. Our novel model ensembling with the text-only GPT-4 consistently improves the model’s performance under all categories, setting the new SoTA performance.

Extra

Xu, Peng, Xiatian Zhu, and David A. Clifton. "Multimodal learning with transformers: A survey." IEEE transactions on pattern analysis and machine intelligence 45.10 (2023): 12113-12132.

Self-Attention for Multi-Modal Interaction

TABLE II
SELF-ATTENTION VARIANTS FOR MULTI-MODAL INTERACTION/FUSION

Self-Attention	Definitions	Streams	Formulations	Complexities	References
Early Summation	token-wise sum before Tfs	1	$\mathbf{Z} \leftarrow Tf(\alpha \mathbf{Z}_{(A)} \oplus \beta \mathbf{Z}_{(B)})$	$\mathcal{O}(N_{(A)}^2)$	[46], [81]
Early Concat.	token sequence concat. before Tfs	1	$\mathbf{Z} \leftarrow Tf(C(\mathbf{Z}_{(A)}, \mathbf{Z}_{(B)}))$	$\mathcal{O}((N_{(A)} + N_{(B)})^2)$	[7], [44], [178], [180]
Hierarchical Att.	2-stream Tfs followed by concat.	$2 \rightarrow 1$	$\mathbf{Z} \leftarrow Tf_3(C(Tf_1(\mathbf{Z}_{(A)}), Tf_2(\mathbf{Z}_{(B)})))$ $C(\mathbf{Z}_{(A)}, \mathbf{Z}_{(B)}) \leftarrow Tf_1(C(\mathbf{Z}_{(A)}, \mathbf{Z}_{(B)})),$	$\mathcal{O}((N_{(A)} + N_{(B)})^2)$	[146],
Hierarchical Att.	early concat. followed by 2-stream Tfs	$1 \rightarrow 2$	$\begin{cases} \mathbf{Z}_{(A)} \leftarrow Tf_2(\mathbf{Z}_{(A)}), \\ \mathbf{Z}_{(B)} \leftarrow Tf_3(\mathbf{Z}_{(B)}). \end{cases}$	$\mathcal{O}((N_{(A)} + N_{(B)})^2)$	[188]
Cross-Attention	exchange query	2	$\begin{cases} \mathbf{Z}_{(A)} \leftarrow MHSA(\mathbf{Q}_B, \mathbf{K}_A, \mathbf{V}_A) \\ \mathbf{Z}_{(B)} \leftarrow MHSA(\mathbf{Q}_A, \mathbf{K}_B, \mathbf{V}_B) \end{cases}$	$\mathcal{O}(N_{(A)}^2)$	[102], [144]
Cross-Att. to Con.	2-stream cross-att. followed by concat.	$2 \rightarrow 1$	$\begin{cases} \mathbf{Z}_{(A)} \leftarrow MHSA(\mathbf{Q}_B, \mathbf{K}_A, \mathbf{V}_A) \\ \mathbf{Z}_{(B)} \leftarrow MHSA(\mathbf{Q}_A, \mathbf{K}_B, \mathbf{V}_B) \end{cases}$ $\mathbf{Z} \leftarrow Tf(C(\mathbf{Z}_{(A)}, \mathbf{Z}_{(B)}))$	$\mathcal{O}((N_{(A)} + N_{(B)})^2)$	[69] [137], [189]

α and β denote weightings. “ATT.”: attention; “CONCAT.”/“CON.”: concatenation; “TFS”: transformer layers. $N_{(A)}$ and $N_{(B)}$ denote the token sequence lengths of two modalities.

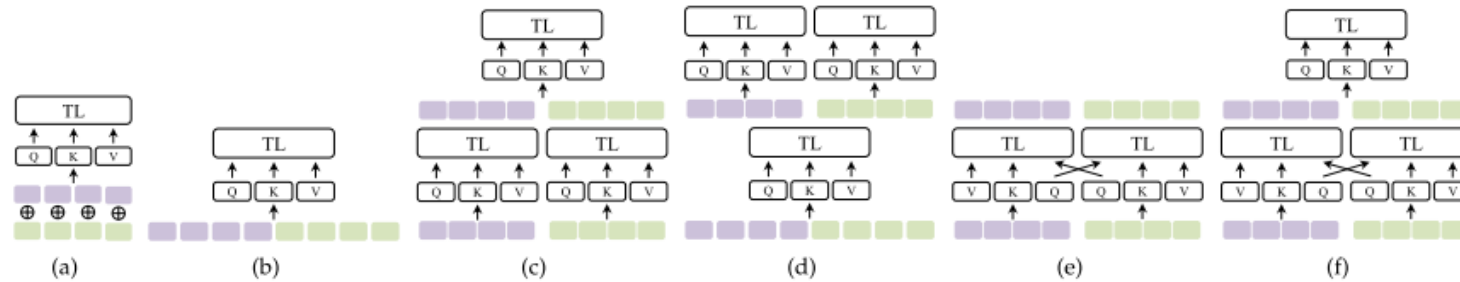


Fig. 2. Transformer-based cross-modal interactions: (a) Early Summation, (b) Early Concatenation, (c) Hierarchical Attention (multi-stream to one-stream), (d) Hierarchical Attention (one-stream to multi-stream), (e) Cross-Attention, and (f) Cross-Attention to Concatenation. “Q”: Query embedding; “K”: Key embedding; “V”: Value embedding. “TL”: Transformer Layer. Best viewed in colour.

Native Multimodal Models

- [1] - Team, Chameleon. "Chameleon: Mixed-modal early-fusion foundation models." *arXiv preprint arXiv:2405.09818* (2024).
- [2] - Cui, Yufeng, et al. "Emu3. 5: Native multimodal models are world learners." *arXiv preprint arXiv:2510.26583* (2025).
- [3] - Team, Qwen. "Qwen3. 5-omni technical report." *arXiv preprint arXiv:2604.15804* (2026).

Native Multimodal Models

Emu 3.5

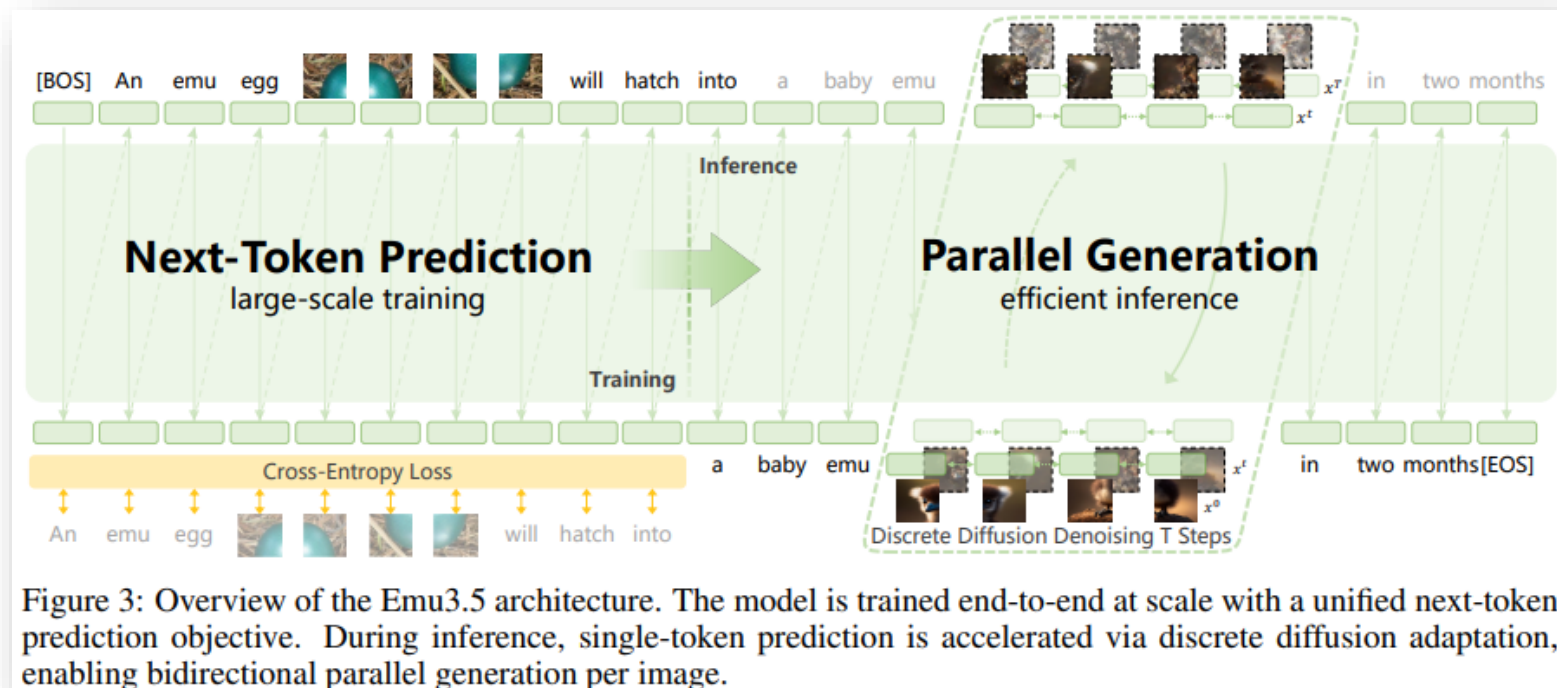


Figure 3: Overview of the Emu3.5 architecture. The model is trained end-to-end at scale with a unified next-token prediction objective. During inference, single-token prediction is accelerated via discrete diffusion adaptation, enabling bidirectional parallel generation per image.

Image encoder/decoder:

Shi, Fengyuan, et al. "Scalable image tokenization with index backpropagation quantization." 2025 IEEE/CVF International Conference on Computer Vision (ICCV). IEEE, 2025.

Native Multimodal Models

Chameleon

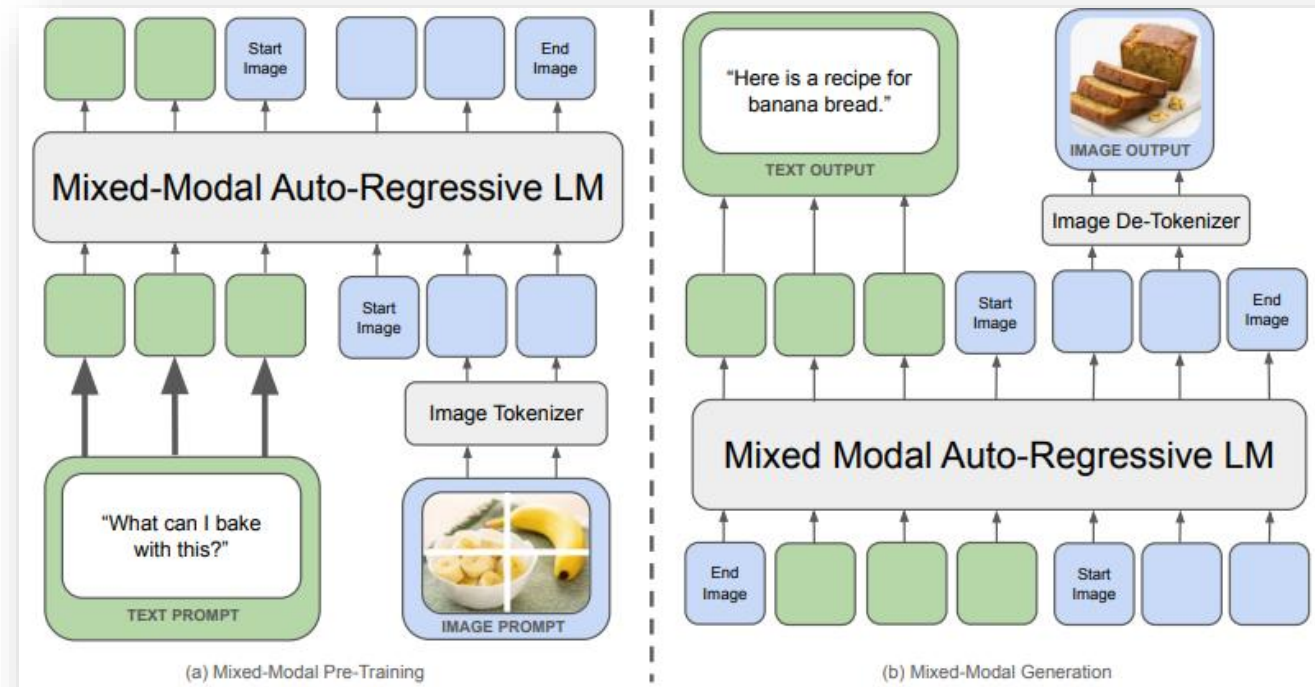
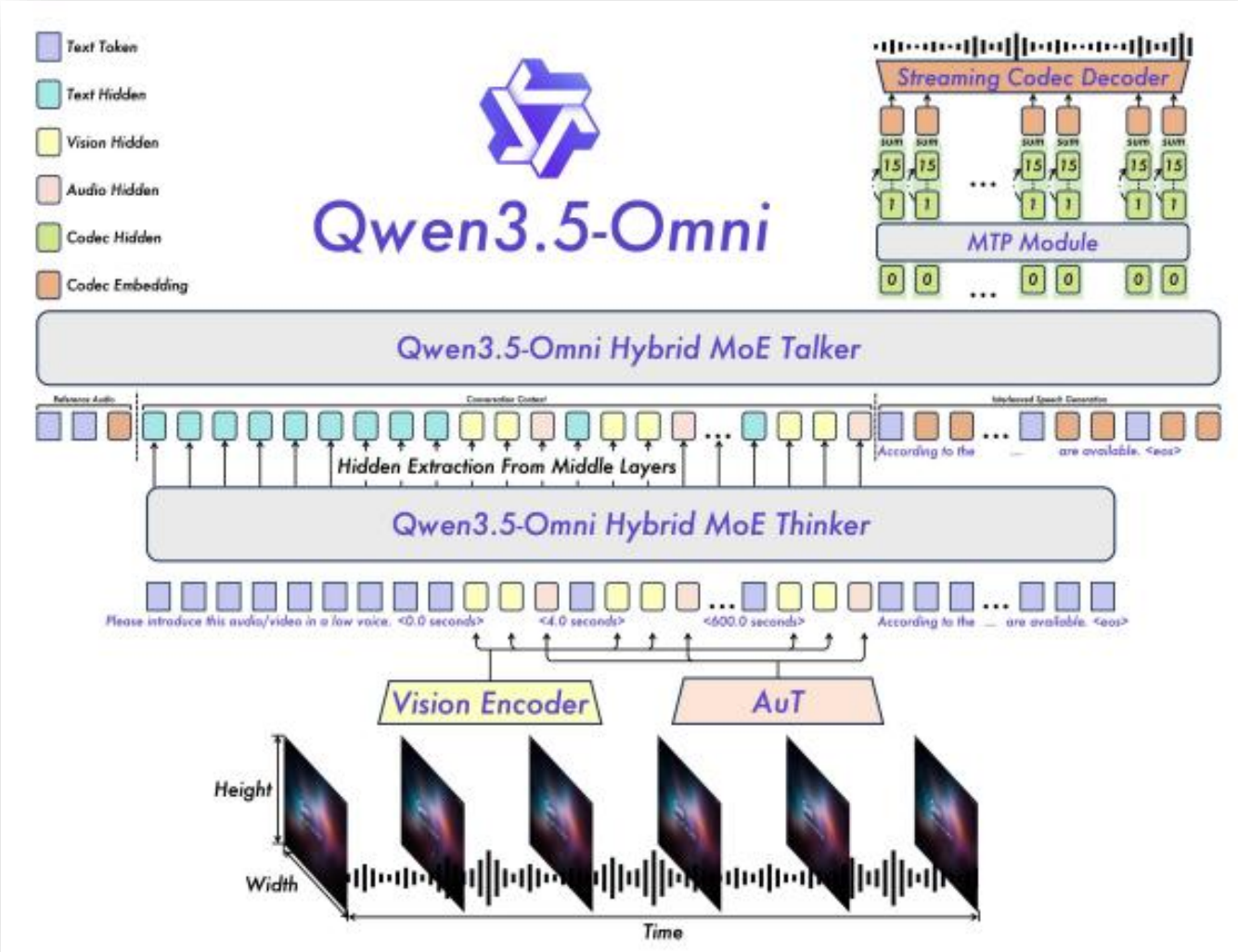


Figure 1 Chameleon represents all modalities — images, text, and code, as discrete tokens and uses a uniform transformer-based architecture that is trained from scratch in an end-to-end fashion on ~10T tokens of interleaved mixed-modal data. As a result, Chameleon can both reason over, as well as generate, arbitrary mixed-modal documents. Text tokens are represented in green and image tokens are represented in blue.

Native Multimodal Models

Qwen 3.5-Omni



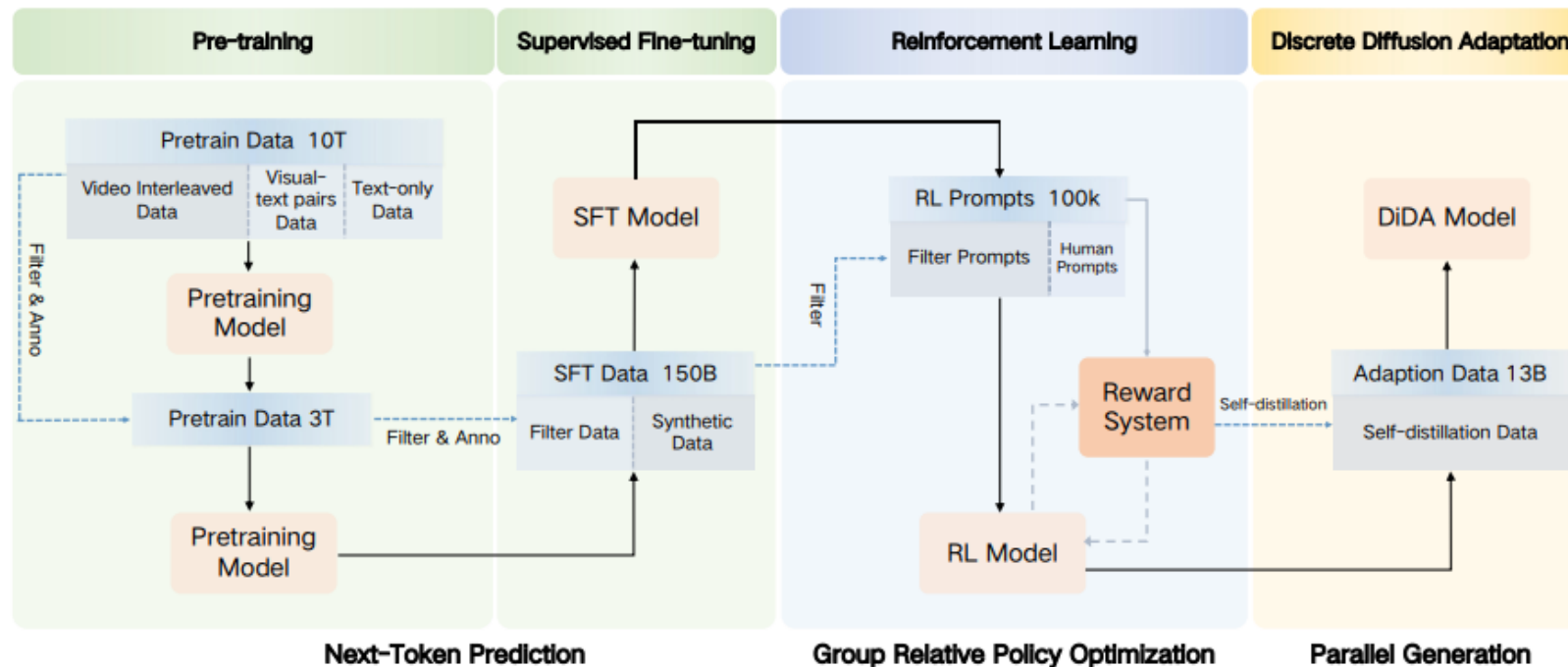
Module	Architecture	Streaming
Audio Encoder	AuT	✓
Vision Encoder	SigLIP2	—
Thinker	Hybrid MoE Transformer	✓
Talker	Hybrid MoE Transformer	✓
MTP	Dense Transformer	✓
Code2wav	ConvNet	✓
First-Packet Latency (Audio Input)		Plus: 435ms Flash: 235ms
First-Packet Latency (Video Input)		Plus: 651ms Flash: 426ms

Native Multimodal Models

Overall Training Pipeline Emu 3.5

Parameters(B)	Layers	Hidden Size	Intermediate Size	Heads (Q / KV)	Vocabulary Size	Context Length
34.1	64	5120	25600	64/8	282926	32768

Table 1: Model configurations of Emu3.5.



Results

	Chameleon		Llama-2			Mistral		Gemini Pro	GPT-4
	7B	34B	7B	34B	70B	7B	8x7B	—	—
Commonsense Reasoning and Reading Comprehension									
PIQA	79.6	83.3	78.8	81.9	82.8	83.0	83.6	—	—
SIQA	57.0	63.3	48.3	50.9	50.7	—	—	—	—
HellaSwag	74.2	82.7	77.2	83.3	85.3	81.3	84.4	—	—
	75.6 10-shot	85.1 10-shot	—	—	87.1 10-shot	83.9 10-shot	86.7 10-shot	84.7 10-shot	95.3 10-shot
WinoGrande	70.4	78.5	69.2	76.7	80.2	75.3	77.2	—	—
Arc-E	76.1	84.1	75.2	79.4	80.2	80.0	83.1	—	—
Arc-C	46.5	59.7	45.9	54.5	57.4	55.5	59.7	—	—
OBQA	51.0	54.0	58.6	58.2	60.2	—	—	—	—
BoolQ	81.4	86.0	77.4	83.7	85.0	84.7*	—	—	—
Math and World Knowledge									
GSM8k	41.6	61.4	14.6	42.2	56.8	52.1 maj@8	74.4 maj@8	86.5 maj@32 CoT	92.0 SFT CoT
	50.9 maj@8	77.0 maj@32	—	—	—	—	75.1* maj@32	—	—
MATH	11.5 maj@1	22.5 maj@1	2.5	6.24	13.5	13.1 maj@4	28.4 maj@4	32.6	52.9**
	12.9 maj@4	24.7 maj@4	—	—	—	—	—	—	—
MMLU	52.1	65.8	45.3	62.6	68.9	60.1	70.6	71.8	86.4

Datasets	Qwen3.5-Plus-Instruct	Qwen3.5-Omni-Flash	Qwen3.5-Omni-Plus
<i>Knowledge</i>			
MMLU-Pro	86.8	79.9	85.9
MMLU-Redux	94.3	90.0	94.2
SuperGPQA	67.4	54.9	66.4
C-Eval	92.3	86.0	92.0
<i>Instruction Following</i>			
IFEval	89.7	85.2	89.7
IFBench	51.1	38.4	52.6
<i>Long Context</i>			
AA-LCR	62.0	46.0	57.0
LongBench v2	60.2	46.4	59.6
<i>STEM</i>			
GPQA	85.9	76.4	83.9
<i>Reasoning</i>			
LiveCodeBench v6	67.1	56.6	65.6
HMMT Nov 25	86.2	59.0	84.4
IMOAnswerBench	68.3	51.5	65.5
<i>General Agent</i>			
BFCL-V4	66.1	55.3	63.3
TAU2Bench	82.7	78.0	81.0

Results

	Model	Model Size	COCO	Flickr30k	VQAv2
Pre-trained	Flamingo-80B	80B	113.8 32-shot	75.1 4-shot	67.6 32-shot
	IDEFICS-80B	80B	116.6 32-shot	73.7 4-shot	65.9 32-shot
Chameleon	Chameleon	34B	120.2 2-shot	74.7 2-shot	66.0 2-shot
	Chameleon-SFT	34B	140.8 0-shot	82.3 2-shot	—
	Chameleon-MultiTask	34B	139.1 2-shot	76.2 2-shot	69.6
Fine-tuned	Flamingo-80B-FT	80B	138.1	—	82.0
	IDEFICS-80B-Instruct	80B	123.2 32-shot	78.4 32-shot	68.8 32-shot
Closed Source (finetuning status unknown)	GPT-4V	—	78.5* 8-shot	55.3* 8-shot	77.2
	Gemini Nano 2	—	—	—	67.5
	Gemini Pro	—	99.8* 2-shot	82.2* 4-shot	71.2
	Gemini Ultra	—	—	—	77.8

Table 7 Model Performances on Image-to-Text Capabilities. * Evaluated using API.

Datasets	Qwen3.5-Plus-Instruct	Qwen3.5-Omni-Flash	Qwen3.5-Omni-Plus
<i>STEM and Puzzle</i>			
MMMU	81.0	76.9	80.1
MMMU-Pro	73.8	68.2	73.9
MathVision	73.6	65.4	73.0
Mathvista (mini)	86.9	82.9	86.1
DynaMath	84.2	79.3	83.8
ZEROBench	6	1	5
ZEROBench_sub	31.1	26.0	34.4
<i>General VQA</i>			
RealWorldQA	79.1	77.5	84.1
MMStar	80.3	75.7	79.4
MMBench _{EN-DEV-v1.1}	93.8	88.8	92.8
SimpleVQA	66.1	54.4	65.3
<i>Text Recognition and Document Understanding</i>			
CharXiv (RQ)	74.2	64.4	72.5
CC-OCR	83.0	80.8	83.4
AI2D_TEST	92.1	89.0	91.2
MMLongBench-Doc	59.7	53.6	57.5
OCRBench	91.4	89.1	91.3
<i>Spatial Intelligence</i>			
ERQA	53.8	50.0	54.8
CountBench	95.1	88.2	95.1
RefCOCO(avg)	95.2	92.6	95.0
ODInW13	50.3	46.8	49.5
EmbSpatialBench	83.4	82.7	85.4
<i>Video Understanding</i>			
VideoMME _(w/o sub.)	81.0	77.0	81.9
MLVU _(M-Avg)	85.1	81.9	86.8
MVBench	76.7	70.8	79.0
LVBench	68.6	65.7	71.2
MMVU	67.1	62.7	67.5
MME-VideoOCR	74.2	70.5	77.0
<i>Medical VQA</i>			
SLAKE	82.8	73.1	84.7
PMC-VQA	62.4	58.7	62.7
MedXpertQA-MM	55.3	44.8	54.7

Results

Datasets	Gemini-3.1 Pro	Qwen3.5-Omni-Flash	Qwen3.5-Omni-Plus
<i>Text Query QA</i>			
DailyOmni	82.7	81.8	84.6
WorldSense	65.5	57.9	62.8
AVUT	85.6	81.4	85.0
AV-SpeakerBench	75.1	65.2	71.3
VideoMME _{w/ audio} ^a	89.0	79.3	83.7
<i>Audio Query QA</i>			
Qualcomm IVD	66.2	66.3	68.5
<i>Caption</i>			
Omni-Cloze	57.2	63.0	64.8
<i>Agent (Tool Use)</i>			
OmniGAIA ^b	68.9	33.9	57.2
^a VideoMME is evaluated with use_audio_in_video=True.			
^b OmniGAIA is evaluated without a thinking prompt and without <answer> formatting. All results are evaluated using DeepSeek-V3.2-Thinking as the judge.			

Results

Chamaleon



Prompt: My mom just adopted a dog called a mountain cur. I thought I knew a lot about dogs but I have never heard of this breed. Can you give me some basic information as well as show me a few pictures of what they look like?

Response: Sure! Mountain Cur dogs are a type of hunting dog that originated in the Appalachian Mountains. They are known for their intelligence, loyalty, and strong prey drive. They are typically medium-sized dogs, with a muscular build and a short, easy-to-maintain coat that comes in a variety of colors including black, brown, tan, red, and white. Here is an image of a black and white mountain cur dog: 

They have a long muzzle, floppy ears, and a bushy tail. They are generally friendly and outgoing, but can be wary of strangers if not socialized properly. They are highly energetic dogs that require regular exercise and mental stimulation to prevent boredom and destructive behavior.

Here is another image of a mountain cur dog in the forest: 

Mountain Cur dogs are often used for hunting, particularly for small game such as squirrels, raccoons, and rabbits. They are also popular family pets due to their loyal and affectionate nature. However, they do require regular exercise and training to ensure they remain happy and healthy. With proper care and attention, Mountain Cur dogs can make wonderful companions for active families or individuals who enjoy the outdoors.



Prompt: What color is a polar bear's fur? Show me a photograph of the polar bear in the wild.

Response: Polar bears have white fur, which helps them blend in with their snowy and icy surroundings. This camouflage allows them to hunt more effectively, as they can sneak up on their prey without being detected. Additionally, the white fur reflects sunlight, keeping the polar bear cool in the harsh Arctic environment. Here is an image of a polar bear in the wild: 

Results

Emu 3.5



Convert to a view as if you are a bird flying above the building



Convert to right side view of the building



Convert to front view of the building



Convert to left side view of the building



Take out your phone and take a photo of the building from top down



Convert to top down view of the building



Shift the perspective to a lower angle, looking up at the hiker as they walk into the forest, emphasizing the towering trees and creating a more dramatic atmosphere.



Move the virtual camera down to show a street-level view

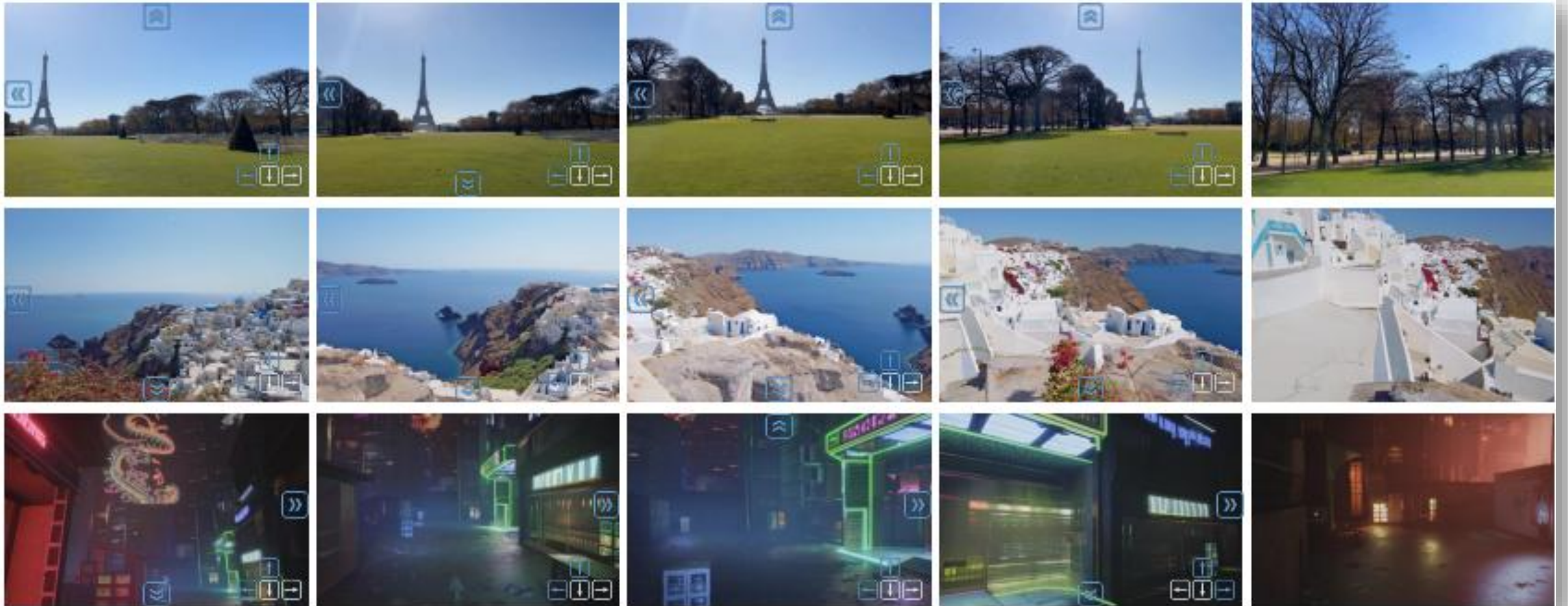


Shift to a gentle bird's-eye view and unlatch the jar so the lid sits open, with a few berries spilled onto the marble.



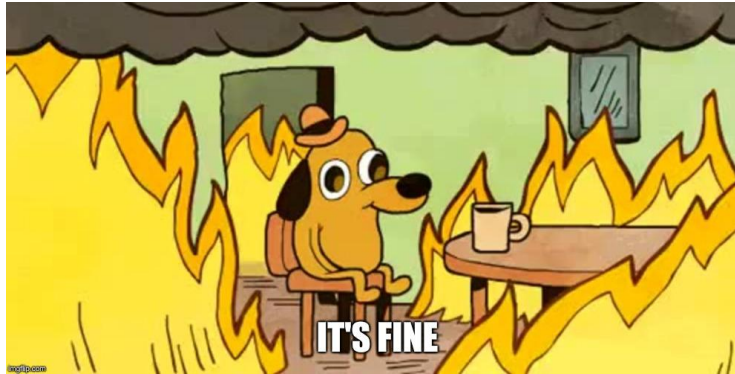
Results

Emu 3.5



Additional References

- [1] - Shukor, Mustafa, et al. "Scaling laws for native multimodal models." *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025.
- [2] - Tong, Shengbang, et al. "Beyond language modeling: An exploration of multimodal pretraining." *arXiv preprint arXiv:2603.03276* (2026).
- [3] - Xiao, Chenghao, et al. "Scaling language-centric omnimodal representation learning." *Advances in Neural Information Processing Systems* 38 (2026): 158370-158401.



Thanks